



Financing strategies and government incentives in a competing supply chain with Trading-Old-for-Remanufactured programs



Peng Ma^{*}, Xuyan Zhou

School of Management Science and Engineering, Nanjing University of Information Science and Technology, Nanjing 210044, PR China

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ABSTRACT

Many supply chain (SC) firms use remanufacturing operations to save costs, but they often face insufficient funds. We consider that remanufacturers with insufficient funds can obtain financing through two methods: bank financing (BF) and retailer financing (RF). This paper considers a SC that implements a Trade-Old-for-Remanufactured (TOR) program, and studies the impacts of financing strategies and the government policy on the SC. By analyzing the optimal solutions of SC and members in different situations, we can find the following results: (i) No competition leads to higher total profits than the competition case. (ii) TOR program will cannibalize new product demand, but it has a positive impact on the total sales volume of new and remanufactured products. (iii) Higher rebate ratio and larger subsidy will increase the total sales quantities of two products and improve the total profits of the SC. (iv) With the same government subsidies, the profit of the remanufacturer under RF strategy is higher than that under BF strategy.

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Introduction

The amount of landfill waste generated across industries has been growing rapidly. For example, in 2020, a number of United Nations agencies jointly released "The Global E-waste Monitoring 2020". According to the report, the world generated a record 53.6 million tons of e-waste in 2019, an increase of 21% in just five years [1]. E-waste contains toxic chemicals such as lead glass, mercury, and cadmium, while it contains abundant resources (e.g., iron, copper, gold), only 17% of the e-waste is properly recycled. E-waste has grown at a rate of more than 5%, partly because of the rise of middle-class families in developing countries, but also because of shortening of product life cycles [2]. When E-waste is landfilled or burned, the metal materials will enter the soil and rivers, which will seriously pollute local land and water resources. After burning, some organic substances will produce a lot of harmful gases, such as dioxins and polychlorinated biphenyls, which are not only harmful to human body, but also enter the atmosphere, causing irreversible consequences to all living things on earth.

In order to promote sustainable environmental development and the recycling of old products, more and more companies are starting

to recycle used products for remanufacturing through reverse logistics. Due to the complexity of the processes involved in the remanufacturing and the extension of the item's life cycle, remanufacturing is the most important key for the sustainable development (Turki et al. [33]; Assid et al. [10]). Many firms are gradually paying attention to the remanufacturing process, and therefore, the Trade-old-For-remanufactured (TOR) programs are introduced by Chinese government in 2013 [22,26,28]. TOR refers to the behavior that consumers who hold used products can buy the remanufactured products by returning their old products and receive the government subsidies [28]. This program is not only beneficial to promote the recycling of used products, but also helps to expand the market of remanufactured products, which is of great significance for firms to create environment-friendly brand image. Because of the many benefits of trade-in program, many firms have carried out this program. For example, many automobile component manufacturers have started providing consumers with trade-in services [3].

Despite remanufacturing has many potential benefits, not all manufacturers are willing to undertake remanufacturing activities due to many reasons. One reason is that many upstream enterprises in the supply chain (SC) are small and medium-sized enterprises (SMEs), which usually have the problem of insufficient capital. A 2019 study made by FED Small Business shows that for American companies, 64% of employer firms with less than 500 employees

^{*} Corresponding author.

E-mail address: mapeng88@126.com (P. Ma).

faced financial challenges in the prior 12 months. Two thirds of companies addressed these challenges by using the owners' personal funds and 54% of small employer firms that applied for \$250 K or less did not receive the full amount of seeking financing [4]. In 2020, COVID-19, which swept the globe, made the financial situation of SMEs become more difficult. According to the data released by China Association of Small and Medium (SME) Enterprises on February 15, 2020, under the impact of COVID-19, the SME development index was 76.4 in February, falling sharply by 16.2 points from the previous month, setting a lowest record in ten years. Among them, the liquidity index and financing index have decreased significantly, and SMEs are facing the risk of capital chain rupture [5]. If the enterprises have financial constraints and business difficulties, they would not choose remanufacturing.

Other reasons that stymie remanufacturing include cannibalization of new product demand by the remanufactured product (abbrev. as "reman" hence forth), brand dilution, uncertain return volumes and yields, and uncertain marketplace acceptance of a reman. In customers' mind, remanufactured products are inferior and their willingness-to-pay (WTP) is lower than the WTP for a new product [14]. Given this resistance from the customer, and the reluctance from the manufacturer, governments are recognizing the importance of their roles for encouraging remanufacturing. A recent report about the UK government investigates the role of remanufacturing. UK's Rt. Hon Dame Caroline Spelman MP, stated "Although opportunities for remanufacturing are far-reaching, Government and industry need to work together to develop its full potential." The report notes "Government should consider implementing a tax break for remanufacturers in order to encourage economic resilience in the UK remanufacturing industry," and "Government should work towards improving and increasing financial investment in the UK remanufacturing industry." Indeed, many governments are also taking action to encourage remanufacturing and other forms of recovery. Recently, India laid down several recycling rules, which establish cash-back incentives and penalize manufacturers who have not implemented effective recycling programs [6].

Many articles have studied the TOR program. Some researchers believe that the implementation of TOR projects can not only benefit enterprises, but also have a beneficial impact on consumers and the environment ([11],[18],[22],[29],[31]). However, these papers do not take into account the impact of the shortage of funds on the TOR project. There are also many papers that consider the capital shortage of firms ([16],[23],[38]). However, these researchers didn't study the problem of fund shortage in the context of TOR. In fact, manufacturers generally face capital constraints and borrow credits from banks or downstream retailers. Most of the above literature studies the TOR program and the SC's financing decisions independently. However, the practical problems will be more complex. For example, the remanufacturers' capital constraints and the government incentives in the TOR program will affect the performance of the SC. In order to fill this gap, this paper will solve the following problems: (i) What are the optimal product prices and profits under each SC scenario? (ii) What are the impacts of remanufacturing subsidies and customer rebates on the optimal prices and the demands of the new product and the reman? (iii) What is the impact of capital constraints on the SC performance? (iv) Which of financing strategies (BF and RF) is better for the remanufacturer?

To solve these problems, we consider a two-stage competitive SC. In addition, we consider two types of government incentives: a remanufacturing subsidy given to the remanufacturer and a rebate given to customer if she purchases a reman. Customers are of two types: first-time customers and replacement customers. Customers in general prefer new products over remans. Both first-time and replacement customers are heterogeneous with respect to their WTP for product quality. The remanufacturer receives a volume

dependent subsidy from the government and a customer receives a rebate if she purchases a reman. The retailers use second-degree price discrimination. We consider (i) when the manufacturer and the remanufacturer cooperate with each other, (ii) when the manufacturer and the remanufacturer compete with each other, (iii) when the remanufacturer receives government subsidies, (iv) When the remanufacturer does not receive government subsidies, (v) When the remanufacturer has capital constraints, and (vi) When the remanufacturer has a sufficient capital. Through the research and comparison of these situations, we draw the following conclusions:

(1) TOR program will cannibalize new product demand, but it has a positive impact on the total sales volume of new and remanufactured products.

(2) Both government subsidies and consumers rebates have a significant impact on TOR program. When manufacturers and remanufacturers are in a competition, the customer rebate increases overall demand and boosts production quantity. And the remanufacturing subsidy always lowers retail prices of new product and reman.

(3) By comparing the total profits of the manufacturer and the remanufacturer, it is found that in most cases, SC profits increase with the government subsidy and the rebate ratio. Also, no competition leads to higher total profits than the competition case.

(4) The financing decision of the remanufacturer will also affect the SC performance. It is found that with the same government subsidies, the profit of remanufacturer under RF strategy is greater than that under BF strategy.

The main contributions of this paper are as follows: (1) Firstly, we consider the remanufacturer with capital constraints and first study the financing decisions of the remanufacturer with TOR program. We assume that the remanufacturer with capital constraints can obtain loans from a bank or the downstream retailer. (2) This research considers the influences of competition and cooperation between the manufacturer and the remanufacturer on the SC performance. (3) We also study the impacts of remanufacturing subsidies and consumer rebates on pricing decisions and SC performance.

Related literature

We divide the remanufacturing literature relevant to our research into three groups: SC competition, SC financing and government subsidies for remanufacturing.

SC competition

Many researchers have studied the competition relationship in the SC, including the competition between manufacturers, remanufacturers, retailers and products. Chen et al. [13] study how to implement price matching when there is competition between a brick-and-mortar retailer and an online retailer. Majumder and Groenevelt [24] study the two-stage game model when OEM and local manufacturer compete, and find that remanufacturing cost has an important impact on product management strategy. Based on a competitive closed-loop SC, He et al. [20] study the recycling efficiency and customer behavior under the inconvenience-perception in collection. Wang et al. [35] study the optimal pricing strategy when the manufacturer and the remanufacturer compete in the recycling market with following three scenarios: recycling rejection, self-implementation collecting and outsourcing collecting. Savaskan et al. [32] study the impacts of forward channel and reverse channel of manufacturers' collection postconsumer products on pricing strategies when retailers are competitive. Wu et al. [36] extend Savaskan's [32] model and examine the impacts of SC competition on the optimal reverse channel choice of manufacturers who remanufacture their own products. Mondal et al. [25] study a SC where

the manufacturer, the retailer, and the recycler engage in the competitive recycling, and design a revenue-and-cost sharing contract. Dey et al. [15] study the SC's operational strategy when there are two competitive retailers selling new product and reman separately. Ma et al. [28] study the effect of subsidies on the SC operations when "trade-old-for-new" and TOR programs co-exist.

The above literature studies the competition of the SC and obtains many meaningful insights. However, most of them do not consider the finance constraints of SC members, and only a few of them consider the government subsidies for remanufacturing.

SC financing

Up to now, more and more researchers begin to pay attention to SC financing. Raghavan et al. [30] study the SC model when the retailer and the manufacturer have capital constraints, and find that the SC can achieve the maximum utility when the borrower provides joint financing. Cai et al. [12] study how a capital-constrained retailer chooses between bank financing and trade financing. Jing et al. [23] study the problem of which financing method should be chosen by the capital-constrained retailer under different circumstances. Yang et al. [38] study how trade credit improves SC efficiency by allowing retailers and suppliers to share part of the demand risk. Tang et al. [34] study two new financing methods, i.e., purchase order financing and buyer direct financing. Yan et al. [37] study the equilibrium financing strategy including bank credit financing and trade credit financing, and analyze the direct relationship between business decision and financial decision. Guo et al. [17] study the implementation of SC finance in coal industry, and find that advance payment mechanism has significant advantages for the SC profit and risk reduction.

Different from the above literature, we consider remanufacturing, the financing decisions of SC enterprises, and also take government subsidies into account. We show that both government subsidies and consumers rebates have significant impacts on remanufacturing.

Government subsidies for remanufacturing

In order to stimulate remanufacturing, many governments have implemented subsidy policies for remanufacturing enterprises, and remanufacturing subsidy has become a current research hotspot. Mitra et al. [27] study the pricing strategy of closed-loop supply chains when government subsidies are distributed between the manufacturer and the remanufacturer. Adda et al. [9] study the short-term and long-term effects of government policies on the trade-in program, and find that subsidies will stimulate the development of the automobile industry in the short term, but may lead to low activity in the long term. Huang et al. [19] study the impacts of government's scrappage program and subsidy on consumers and manufacturers in the automobile SC. Han et al. [22] study the impacts of many factors on the TOR program, including receptivity, durability and subsidy, and find that high government subsidies can encourage enterprises to implement TOR projects. Ma et al. [26] study the impacts of double reference effects on pricing decisions and the manufacturer's performance with government subsidies.

Above all, although there are many papers focusing on government subsidies for remanufacturing, few researchers pay attention to the financial constraints of upstream firms and the SC competition. In this paper, we provide a more practical model by considering various factors, such as customer rebate ratio and the customer's WTP, then get the optimal solutions of SC members in different situations.

We choose some classical literature related to our research to highlight the contributions of our study (see Table 1).

Model description

We consider a two-stage competitive SC. It should be noted that in this paper, the competitive SC means that the manufacturer sells the new product through retailer 1 and the remanufacturer sells the reman through retailer 2 (See Fig. 1). The retailers buy their respective products at wholesale prices those are set by the manufacturer and the remanufacturer. The retailers then sell their respective products to the customers. There are two categories of customers: First-time customers and replacement customers. As the name indicates, first-time customers buy the product for the first time, and replacement customers own a previously purchased product that they can trade in. Both first-time and replacement customers can purchase either the new or the reman, but perceive the new product to be of higher quality. Replacement customers can trade in their old product or sell it in the secondary market. We aim to investigate the effect of government policy, including (i) offering the customer rebate for buying the reman and (ii) giving the remanufacturing subsidy to the remanufacturer for producing the reman. The former incentive makes a reman more attractive to the end user, while the latter subsidy motivates the remanufacturer to institute remanufacturing processes. Together, they affect product prices, customer demand, and SC profits. Fig. 1 shows this two-stage competitive SC, in which new products are produced by the manufacturer and sold by Retailer 1, while the reman is made by the remanufacturer and sold by Retailer 2.

We make the following assumptions:

Assumption 1. This paper considers a two-stage competitive SC with one manufacturer, one remanufacturer, and two retailers selling new product and reman, respectively.

Assumption 2. The quality of new products is higher than that of the reman. Based on this, we assume that the consumer's WTP for a reman is a fraction γ ($0 < \gamma < 1$) of that of the new product. And the WTP of customers for new product is denoted by θ , which is uniformly distributed between 0 and 1.

Assumption 3. The product demand depends on the retail price, and there are sufficient resources to manufacture the products to meet the demand. Thus, demand, sales quantity and production quantity are all equal.

Assumption 4. The government can provide two types of policy incentives: remanufacturing subsidies and consumer rebates.

Assumption 5. Capital-constrained remanufacturers can obtain financing from a bank or the downstream retailer, called bank financing (BF) and retailer financing (RF) respectively.

Assumption 6. We assume that the total market size is 1, replacement consumers comprise a fraction λ of this market and new consumers comprise $1 - \lambda$.

Assumption 7. Replacement consumers receive an amount τ per unit if they trade in their pre-owned product for reman, and an amount v per unit if they sell their old products in the secondary market (and replace them with new products). Clearly,

$\tau \geq v$, else the customers would not have any incentive to trade in their pre-owned products from the remanufacturer.

Table 2 summarizes our parameters and decision variables.

Consider a first-time customer with valuation θ . When the government does not offer a rebate, this customer's utility is $\theta - p_n$ for a new product and $\gamma\theta - p_r$ for a remanufactured product. The replacement customer derives some value from her old product, but this value depends on whether she trades in the old product (if she buys a reman) or sells it in the secondary market (buys a new product). Therefore, the utility for this customer is $\theta - p_n + v$ if she buys a new product, and $\gamma\theta - p_r + \tau$ if she buys a reman. Similarly, we can

Table 1
(a). Summary of the main literature on SC Competition (Section 2.1).

Authors	SC Competition	SC Financing	Government subsidies for remanufacturing	Focus	SC structure & Model
Chen et al.[13]	✓			Price matching	Duopoly, Hotelling model
Majumder and Groeneveld[24]	✓			Impact of remanufacturing costs on the management	1-OME & 1-local remanufacturer; two -period model
He et al.[20]	✓			Competitive collection	Two-echelon decentralized model
Wang et al.[35]	✓			Competitive recycling- market	Closed-loop supply chain& Stackelberg gaming models
Savaskan et al.[32]	✓			Reverse channel & forward channel	1-manufacturer& 1-retailer
Wu et al.[36]	✓			Optimal choice of reverse channel	1-manufacturer & 2-retailer, multiechelon distribution model
Mondal et al.[25]	✓			Pricing and collection strategy in a CLSC	1-manufacturer, 1-retailer and 1- third-party collector
Dey et al.[15]	✓			Two competitive retailer's different game strategies	1-manufacturer, 2-suppliers and 2- retailers
Ma et al.[28]	✓			"Trade old for new" (TON) and TOR programs	Closed-loop supply chain
Table 1(b). Summary of the main literature on SC Financing (Section 2.2)					
Raghavan et al.[30]	✓			Both the retailer and manufacturer are facing financial constraints	1-retailer & 1-manufacturer, two -level SC
Cai et al.[12]	✓			Trade credit & bank credit	1-capital-constrained retaile, 1- manufacturer&a bank
Jing et al.[23]	✓			Trade credit & bank credit	1-retailer, 1- manufacturer&a bank
Yang et al.[38]	✓			The risk- sharing role of trade credit	1-retailer & 1-manufacturer
Tang et al.[34]	✓			Purchase order financing (POF)&buyer direct financing (BDF)	1-manufacturer, 1-financially constrained supplier& 1-bank
Yan et al.[37]	✓			Partial credit guarantee (PCG) contract	1-capital-constrained retailer, 1-manufacturer& 1- commercial bank
Guo et al.[17]	✓			The advance payment mechanism	1-coal company, 1-electricity company&1-bank
Table 1(c)- Summary of the main literature on Trade-in programs in government subsidies for remanufacturing (Section 2.3)					
Mitra et al.[27]		✓	✓	The: Effects of government subsidies	1-manufacturer & 1- remanufacturer; two-period model
Adda et al.[9]		✓	✓	The effects of subsidies on durable goods markets	Policy functions
Huang et al.[19]		✓	✓	Automobile scrappage programs	Automobile supply chain involving 1-manufacturer&1-retailer
Ma et al.[26]			✓	Double reference effects	1-manufacturer;two-pcrid model
Han et al.[22]			✓	Receptivity, durability, and subsidy	Game theory
This Research	✓	✓	✓	Pricing & SC financing, heterogeneous customers	1-manufacturer, 1- remanufacturer, & 2-retailer; Game Theory

determine the customer utilities when government does provide an incentive. Table 3 gives the utility functions of the first-time and the replacement customers with and without incentives.

We first consider the case where there is no capital constraint. Section 3.1 analyzes the case when there are no government incentives, while Section 3.2 considers that with government incentives. Section 3.3 focuses on the situation where there are financial restrictions on the remanufacturer and no government subsidies. Section 3.4 pays attention to the situation when the remanufacturer has capital constraints with government subsidies.

Under Sections 3.1 and 3.2, we look at two cases: when the manufacturer and the remanufacturer cooperate, and when they compete with each other. Under Sections 3.3 and 3.4, we only consider the competitive situation. The manufacturer and the remanufacturer set wholesale prices based on their retailers' response functions. The retailers respectively set their retail prices to maximize profits.

No government incentive programs and no capital constraints

Manufacturer and remanufacturer cooperate (Scenario 000)

We now consider the situation when the remanufacturer has sufficient capital, the government does not have any incentive programs, the manufacturer and the remanufacturer cooperate. Effectively, we can assume that one entity, say the manufacturer, makes both the new and the remanufactured products. An example of this scenario is in the hospital bed industry, where Hill-Rom sells new products. The company also takes back used equipment, and sells remanufactured products in the same market (e.g., [18]). In this case, the following utility relationships (Table 3) are satisfied for a first-time customer who buys a new product:

$$U_n^a > U_r^a \Rightarrow \theta > \frac{p_n - p_r}{1 - \gamma}, \text{ and } U_n^a > 0 \Rightarrow \theta > p_n$$

The first condition states that the first-time customer's utility from the new product exceeds the first-time customer's utility from the replacement product. The second condition states that the first-time customer's utility from the new product is positive. These conditions follow since the customer chooses to buy the new product over the reman.

Note that if $p_n \leq \frac{1}{\gamma}p_r$, then all customers (weakly) prefer the new product to the reman, and so the reman will not be offered. On the other hand, when $p_n > \frac{1}{\gamma}p_r$, the first-time consumer's WTP for the

new product lies in the range $\left[\frac{p_n - p_r}{1 - \gamma}, 1\right]$. Now consider the replacement customer. When $U_n^b > U_r^b$ and $U_n^b > 0$, the replacement consumer's WTP for the new product lies in the range $\left[\frac{p_n - p_r + \tau - v}{1 - \gamma}, 1\right]$ if $p_n - v > \frac{p_r - \tau}{\gamma}$. Thus,

$$\max_{p_n} \Pi_{R1}^{000}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma}\right](p_n - w_n),$$

$$\max_{p_n} \Pi_{R1}^{000}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma}\right](p_n - w_n),$$

$$\max_{p_n} \Pi_{R1}^{000}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma}\right](p_n - w_n),$$

$$\text{and } \max_{p_n} \Pi_{R1}^{000}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma}\right](p_n - w_n).$$

Therefore, the total new product demand, $\max_{p_n} \Pi_{R1}^{000}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma}\right](p_n - w_n)$ is

$$\max_{p_n} \Pi_{R1}^{000}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma}\right](p_n - w_n),$$

and the total reman demand, q_r is

$$\max_{p_n} \Pi_{R1}^{000}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma}\right](p_n - w_n).$$

To determine the equilibrium solution for the competitive SC, we will solve the two-stage, non-cooperative game using backward induction. We begin by characterizing the best-response functions of the two retailers. Given the wholesale price of new product (w_n) and the wholesale price of the reman (w_r), the objectives of the two retailers are:

$$\max_{p_n} \Pi_{R1}^{000}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma}\right](p_n - w_n) \quad (1)$$

and

$$\begin{aligned} \max_{w_r, w_n} \Pi_M^{000}(w_n, w_r) = & q_r^a(w_n, w_r)(w_r - c_r) \\ & + q_r^b(w_n, w_r)(w_r - c_r - \tau) + q_n(w_n, w_r)(w_n - c_n) \end{aligned} \quad (2)$$

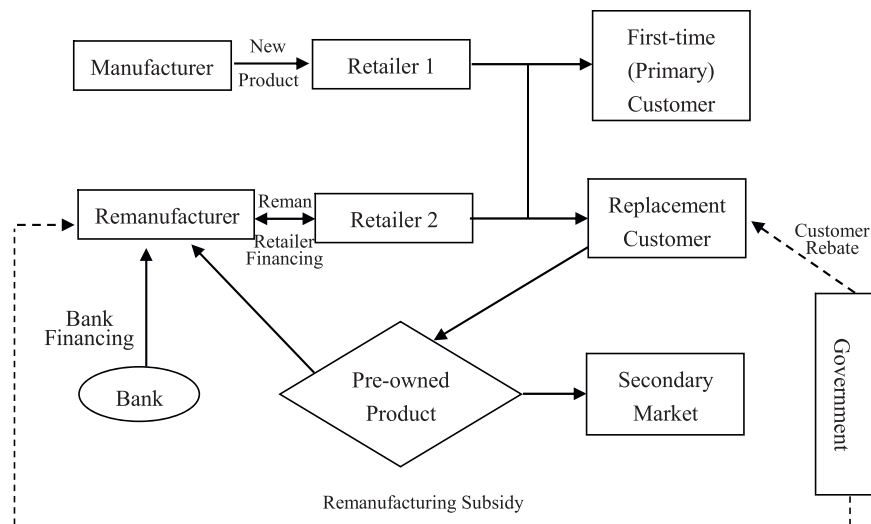


Fig. 1. The Competitive SC model.

Table 2
Notation Summary.

Parameters	
Scenario i, j, k	Index $i = 0$ denotes no government incentives , $i = 1$ denotes with incentives;
q_r^{ijk*}	Index $j = 0$ denotes no competition (i.e., cooperation), $j = 1$ denotes competition
q_r^{ijk*}	Index $k = 0$ denotes no financing , $k = 1$ denotes bank financing , $k = 2$ denotes retailer financing
c_n	Customer k 's utility function for the new product, where $k = a$ ($k = b$) corresponds to the first-time (replacement) customer
c_r	Customer k 's utility functions for the reman, where $k = a$ ($k = b$) corresponds to the first-time (replacement) customer
γ	Unit cost of manufacturing the new product
θ	Unit cost of making the remanufactured product
s	Customer receptivity for a reman. When $\gamma = 1$, customer WTP for the reman equals the WTP for the new product (that is, customers consider the reman quality comparable to the new product quality)
ρ	Customer valuation of the new product
λ	Subsidy received by the remanufacturer for making one unit of the reman
τ	Customer rebate ratio, which equals <i>customer rebate/reman price</i>
v	Proportion of replacement customers
r_B	Unit trade-in value of the old product given by the remanufacturer
r_R	Unit salvage value that a replacement customer gets in the secondary market for the old product
Decision variables	
p_n	Unit retail price of the new product
p_r	Unit retail price of the reman
w_n	Unit wholesale price of the new product
w_r	Unit wholesale price of the remanufactured product
Other notation	
q_r^{ijk*}	Reman demand from customer k , where $k = a$ ($k = b$) corresponds to the first-time (replacement) customer
q_r^{ijk*}	New product demand from customer k , where $k = a$ ($k = b$) corresponds to the first-time (replacement) customer
q_r	Total reman demand which equals q_r^{ijk*} . (This also equals the total reman produced and sold.)
q_n	Total new product demand which equals q_r^{ijk*} . (This also equals the total new product produced and sold.)
q_r^{ijk*}	Optimal reman demand for different scenarios
q_n^{ijk*}	Optimal new product demand for different scenarios
p_n^{ijk*}	Optimal unit price of the new product for different scenarios
p_r^{ijk*}	Optimal unit price of the remanufactured product for different scenarios
Π_{R1}^{ijk}	Retailer 1's profit from selling new products for different scenarios
Π_{R2}^{ijk}	Retailer 2's profit from selling reman for different scenarios
Π_M^{ijk}	Manufacturer's profit for different scenarios
Π_{RM}^{ijk}	Remanufacturer's profit for different scenarios

Table 3
Customer utilities under incentive and no incentive cases.

No incentive	First-time Customer	Replacement Customer
New product	$p_n \leq \frac{1}{\gamma} p_r$	$p_n \leq \frac{1}{\gamma} p_r$
Reman	$p_n \leq \frac{1}{\gamma} p_r$	$p_n \leq \frac{1}{\gamma} p_r$
Customer rebate and Remanufacturing subsidy	First-time Customer	Replacement Customer
New product	$p_n \leq \frac{1}{\gamma} p_r$	$p_n \leq \frac{1}{\gamma} p_r$
Reman	$p_n \leq \frac{1}{\gamma} p_r$	$p_n \leq \frac{1}{\gamma} p_r$

Since the manufacturer will decide on the wholesale prices of the new product and the reman, the objective function of the manufacturer is:

$$\max_{w_r, w_n} \Pi_M^{000} \left(w_n, w_r \right) = q_r^a \left(w_n, w_r \right) \left(w_r - c_r \right) + q_r^b \left(w_n, w_r \right) \left(w_r - c_r - \tau \right) + q_n \left(w_n, w_r \right) \left(w_n - c_n \right). \quad (3)$$

We can get the optimal equilibrium solution of the Stackelberg game between the retailers and the manufacturer. **Theorem 1** summarizes our findings.

Theorem 1. *The optimal retail prices, the optimal quantities of the new product and the reman, the optimal wholesale prices of the new product and the reman, and the optimal total profits of manufacturer and remanufacturer are given in the online supplements.*

Proof. Please see the Online Supplement.

Manufacturer and remanufacturer compete (Scenario 010)

Scenario 010 can be found in Caterpillar, the US construction machinery giant, with globally located remanufacturing factories. Caterpillar not only remanufactures its own product, but also collects and remanufactures products of other manufacturers (e.g., [21]). Here, the manufacturer produces the new product while the remanufacturer produces the reman. Without the government incentive, the retailers' reaction functions are the same as those in Scenario 000, i.e., Eqs. (A1) and (A2). Given these reaction functions, the remanufacturer and the manufacturer make decisions independently and concurrently, and need to solve the following optimization problems to maximize their respective profits:

$$\max_{w_r, w_n} \Pi_{RM}^{010} (w_r) = q_r^{a010*} \left(w_n, w_r \right) \left(w_r - c_r \right) + q_r^{b010*} \left(w_n, w_r \right) \left(w_r - c_r - \tau \right) = \frac{(-w_r + c_r)(\lambda - 1)((-\gamma^2 + \gamma)\lambda(\tau - v) + \lambda(\tau - 2v)\gamma^2 + ((-\tau + 4v)\lambda + w_n + w_r)\gamma - 2\lambda\tau - 2w_r)}{(\gamma - 4)(\gamma - 1)\gamma} \quad (4)$$

and

$$\max_{w_r, w_n} \Pi_M^{010} (w_n) = q_n^{010*} \left(w_n, w_r \right) \left(w_n - c_n \right) = \frac{(-w_n + c_n)(\gamma^2 + (-\lambda v + w_n - 5)\gamma + (-\tau + 2v) - 2w_n + w_r + 4)}{(\gamma - 4)(\gamma - 1)} \quad (5)$$

The following theorem summarizes the optimal solutions for the manufacturer and remanufacturer competing case.

Theorem 2. *The optimal retail prices, the optimal quantities of the new product and the reman, and the optimal wholesale prices of the new product and the reman, the profits of remanufacturer and manufacturer are given in the online supplements.*

Proof. Please see the Online Supplement.

With customer rebate and remanufacturing subsidy and without capital constraints

We now consider the case of joint customer rebates and remanufacturing subsidies as seen in China's TOR policy and we consider that the remanufacturer has a sufficient capital. In the US, tax incentives are often given to remanufacturers who set up remanufacturing operations. For example, the State of Missouri gave incentives to John Deere Reman for setting up a facility in Strafford

[7]. This subsection considers the situation when the remanufacturer gets a reman subsidy, and the consumer gets a rebate, which then affects both the supply and demand (and, consequently prices). As in Section 3.1, we study two scenarios, i.e., cooperation and competition between the manufacturer and the remanufacturer.

Manufacturer and remanufacturer cooperate (Scenario100)

To determine the equilibrium solutions using backward induction, we first assume that the wholesale prices are given. Given the wholesale prices, we analyze the retailers' problem to determine the optimal prices and quantities. We then use these results to determine the optimal wholesale prices. Similar to §3.1.1, when the manufacturer and remanufacturer cooperate, we assume they are one entity.

The first-time consumer will purchase a new product from Retailer 1 if $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$ and $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$, while she will purchase the reman from Retailer 2 if $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$ and $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$. Likewise, the replacement consumer purchases a new product from Retailer 1 if $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$ and $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$, while he will purchase a reman from Retailer 2 if $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$. From the above, we find.

$\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$,
and $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$.
If $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$, the first-time consumer's WTP for the new product is in the range of $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$. When $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$ and $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$, we find that the replacement consumer's WTP for the new products is in the range.

$\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$
if $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$.
Thus,
 $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$
 $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$
 $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$ and.
 $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$

Then, the production quantities of the new product and the reman are:

$$\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n) \quad (6)$$

and

$$\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n) \quad (7)$$

As in the previous section, we use backward induction to derive the equilibrium solutions for the competitive SC, by first characterizing the best-response functions of the retailers. Given $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$ and $\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n)$, the retailers' problems become

$$\max_{p_n} \Pi_{R1}^{100}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}\right)(p_n - w_n) \quad (8)$$

and

$$\max_{p_r} \Pi_{R2}^{100}(p_r) = \left(\frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma} - \frac{p_r - \rho p_r - \lambda\tau}{\gamma}\right)(p_r - w_r) \quad (9)$$

The manufacturer's problem, with the objective of maximizing the total profit from selling the new and remanufactured products, can be expressed as

$$\max_{w_r, w_n} \Pi_M^{100} = q_r^a \left(w_r - c_r + s\right) + q_r^b \left(w_r - c_r - \tau + s\right) + q_n \left(w_n - c_n\right). \quad (10)$$

The following theorem summarizes both SC members' optimal solutions.

Theorem 3. The optimal wholesale prices, the optimal production quantities (market demands) of the new products and the reman, the optimal retail prices, optimal wholesale prices, and optimal total profits of manufacturer and remanufacture are given in the online supplements.

Proof. See Online Supplements.

Manufacturer and remanufacturer compete (Scenario 110)

The retailers' reaction functions are the same as those in Section 3.2.1. We thus have

$$\begin{aligned} q_n^{110} &= 1 \\ &= \frac{\gamma^2 + \left(\lambda v - w_n^{110} - 3\right)\gamma + \left(\tau - 2v\right)\lambda + 2w_n^{110} + 2 + \left(-1 + \rho\right)w_r^{110}}{(-1 + \gamma)(\gamma - 4)} \\ &\quad - \frac{(-1 + \lambda)((\lambda v + 1)\gamma^2 + ((-1 + \rho)w_r^{110} - 3\lambda v - w_n^{110} - 1)\gamma + (-2\rho + 2)w_r^{110} + 2\lambda\tau)}{\gamma(-1 + \gamma)(\gamma - 4)}, \\ q_r^{a110} &= \frac{\lambda((\lambda v - v + 1)\gamma^2 + ((-1 + \rho)w_r^{110} - 3\lambda v + \tau + 4v - w_n^{110} - 1)\gamma + (-2\rho + 2)w_r^{110} + 2\tau(\lambda - 2))}{\gamma(-1 + \gamma)(\gamma - 4)}, \\ q_r^{b110} &= \frac{\gamma\lambda v + \rho w_r^{110} + \lambda\tau - 2\lambda v + 2\gamma - 2w_n^{110} - w_r^{110} - 2}{\gamma - 4} \\ p_n^{110} &= \frac{\gamma\lambda v + \rho w_r^{110} + \lambda\tau - 2\lambda v + 2\gamma - 2w_n^{110} - w_r^{110} - 2}{\gamma - 4} \\ p_r^{110} &= \frac{\gamma\lambda\tau + \gamma\lambda v + \gamma^2 - \gamma w_n^{110} + 2\rho w_r^{110} - 2\lambda\tau - \gamma - 2w_r^{110}}{\gamma\rho - \gamma - 4\rho + 4} \end{aligned}$$

When the manufacturer and remanufacturer compete, they set the wholesale prices to maximize their individual profits. To obtain the optimal wholesale prices of the two products, we solve the following two optimization problems:

$$\begin{aligned} \max_{w_n} \Pi_M^{110} &= q_n^{110*} \left(w_n^{110} - c_n \right) \\ &= \left(1 - \frac{\gamma^2 + (\lambda v - w_n^{110} - 3)\gamma + (\tau - 2v)\lambda + 2w_n^{110} + 2 + (-1 + \rho)w_r^{110}}{(-1 + \gamma)(\gamma - 4)} \right) \\ &\quad \left(w_n^{110} - c_n \right) \end{aligned} \quad (11)$$

$$\max_{w_r} \Pi_{RM}^{110} = q_r^{a110} \left(w_r^{110} - c_r + s \right) + q_r^{b110} \left(w_r^{110} - c_r - \tau + s \right) \quad (12)$$

We can derive the optimal equilibrium solutions for SC members. **Theorem 4** summarizes our findings.

Theorem 4. When customers receive a rebate and remanufacturers receive a remanufacturing subsidy and the manufacturer compete with the remanufacturer, the optimal wholesale price, the optimal retail price, optimal products demand, optimal profits of manufacturer and the optimal profits of remanufacturer are given in the online supplements.

Proof. See Online Supplements.

Without government incentive and remanufacturer faces capital constraint

The ways of financing will have significant impacts on the remanufacturer's decision making and production program. In this section, we consider the remanufacturer has an initial capital B, with $B < c_r q_r$, so the remanufacturer needs to borrow money from others. We assume the remanufacturer can get financing from a bank or the retailer and we only consider the competitive situation. In addition, we have some new assumptions:

- (i) The remanufacturer has an insufficient initial capital and a limited liability;
- (ii) The interest rate is exogenous and relatively stable;
- (iii) The retailer has enough money and can lend money to the remanufacturer;
- (iv) In order to eliminate the risk of bankruptcy, we assume $q_r \geq \frac{(c_r q_r - B)(1 + r_B)}{w_r}$

Bank financing (Scenario 011)

Such as Scenario 000, we can get the total market demand of new product and the reman.

$$\max_{p_n} \Pi_{R1}^{011}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma} \right] (p_n - w_n) \quad (13)$$

$$\max_{p_n} \Pi_{R1}^{011}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma} \right] (p_n - w_n) \quad (14)$$

$$\max_{p_n} \Pi_{R1}^{011}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma} \right] (p_n - w_n) \quad (15)$$

$$\max_{p_n} \Pi_{R1}^{011}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma} \right] (p_n - w_n) \quad (16)$$

The retailers' profit functions can be written as:

$$\max_{p_n} \Pi_{R1}^{011}(p_n) = \left[1 - \frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma} \right] (p_n - w_n) \quad (17)$$

$$\max_{p_r} \Pi_{R2}^{011}(p_r) = \left[\frac{p_n - p_r + \lambda(\tau - v)}{1 - \gamma} - \frac{p_r - \lambda\tau}{\gamma} \right] (p_r - w_r) \quad (18)$$

Since we suppose the manufacturer and remanufacturer are competitive and the remanufacturer can borrow money from a bank, the profit functions of the manufacturer and the remanufacturer can be written as:

$$\max \Pi_M^{011}(w_n, w_r) = q_n(w_n^{011} - c_n) \quad (19)$$

$$\begin{aligned} \max \Pi_{RM}^{011}(w_n, w_r) &= q_r^{a011}(w_r^{011} - c_r) + q_r^{b011}(w_r^{011} - c_r - \tau) - (c_r q_r^{011} - B) \left(1 + r_B \right) \\ &\quad (20) \end{aligned}$$

We can get the optimal solutions of the retailers, the manufacturer and the remanufacturer.

Theorem 5. The optimal production quantities (market demands) of the new products and the reman, the optimal retail prices and the wholesale prices, the optimal profits of manufacturer and remanufacturer are given in the online supplements respectively.

Proof. See Online Supplements.

Retailer financing (Scenario 012)

In addition to financing from a bank, a wealthy retailer, as a member of the SC, may lend money to the capital-constrained remanufacturer. In this section, we discuss how the two retailers, the manufacturer and the remanufacturer make decisions to maximize their profits when the remanufacturer needs to borrow money from the retailer and the manufacturer and the remanufacturer are competitive. At time zero, the retailer sets the interest rate r_R and lends $c_r q_r - B$ to the remanufacturer. At the end of the period, the remanufacturer repays the loans to the retailer.

Same as 3.3.1, we have the total market demand of the new product and the reman.

$$\max \Pi_{R1}^{012}(p_n) = q_n(p_n - w_n)$$

$$\max \Pi_{R1}^{012}(p_n) = q_n(p_n - w_n)$$

$$\max \Pi_{R1}^{012}(p_n) = q_n(p_n - w_n)$$

$$\max \Pi_{R1}^{012}(p_n) = q_n(p_n - w_n)$$

Since the retailer can lend money to the remanufacturer, the two retailers' profit functions can be written as follows:

$$\max \Pi_{R1}^{012}(p_n) = q_n(p_n - w_n) \quad (21)$$

$$\max \Pi_{R2}^{012}(p_r) = q_r(p_r - w_r) + (q_r c_r - B) \left(1 + r_R \right) \quad (22)$$

Because the manufacturer and remanufacturer are competitive and the remanufacturer can borrow money from the retailer, the profit functions of the manufacturer and the remanufacturer can be written as:

$$\max \Pi_M^{012}(w_n, w_r) = q_n(w_n^{012} - c_n) \quad (23)$$

$$\begin{aligned} \max \Pi_{RM}^{012}(w_n, w_r) \\ = q_r^{a012}(w_r^{012} - c_r) + q_r^{b012}(w_r^{012} - c_r - \tau) - (c_r q_r^{012} - B) \left((1 + r_R) \right) \end{aligned} \quad (24)$$

We can get the optimal equilibrium solution between the retailers, the manufacturer, and the remanufacturer.

Theorem 6. When the remanufacturer received a loan from the retailer, the optimal retail prices, the optimal production quantities (market demands) of the new products and the reman, the optimal wholesale price, the optimal profits of manufacturer and remanufacturer are given in the online supplements.

Proof. See Online Supplements.

With customer rebate and remanufacturing subsidy and the remanufacturer faces capital constraint

In this section, we suppose the government subsidizes remanufactured products and the remanufacturer offers consumers a rebate if they trade in old products for new ones. In addition, we assume that the manufacturer and the remanufacturer are competitive. Since the remanufacturer's initial capital B is smaller than $c_r q_r$, it can borrow credits from a bank or the retailer. Under the above background, we carried out the following research.

Bank financing (Scenario 111)

Similar to 3.2.1, we can get the total demand of new products and the reman:

$$q_r^b = \lambda \left(\frac{p_n - p_r + \tau - v + \rho p_r}{1 - \gamma} - \frac{p_r - \tau - \rho p_r}{\gamma} \right)$$

$$q_r^b = \lambda \left(\frac{p_n - p_r + \tau - v + \rho p_r}{1 - \gamma} - \frac{p_r - \tau - \rho p_r}{\gamma} \right)$$

$$q_n = q_n^a + q_n^b = 1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma},$$

$$q_n = q_n^a + q_n^b = 1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}.$$

As in the previous section, we use backward induction to derive the equilibrium solutions for the competitive SC. Given w_n and w_r , the retailers' problems become:

$$\max_{p_n} \Pi_{R1}^{111}(p_n) = \left(1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma} \right) (p_n - w_n) \quad (25)$$

$$\begin{aligned} \max_{p_r} \Pi_{R2}^{111}(p_r) = & \left(\frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma} \right. \\ & \left. - \frac{p_r - \rho p_r - \lambda\tau}{\gamma} \right) (p_r - w_r) \end{aligned} \quad (26)$$

When the remanufacturer receives subsidies from government and the remanufacturer can borrow money from a bank. To obtain the optimal solutions of the SC, we solve the following two optimization problems:

$$\max \Pi_M^{111}(w_n, w_r) = q_n(w_n^{111} - c_n) \quad (27)$$

$$\begin{aligned} \max \Pi_{RM}^{111}(w_n, w_r) \\ = q_r^{a111}(w_r^{111} - c_r + s) + q_r^{b111}(w_r^{111} - c_r - \tau + s) - (c_r q_r^{111} - B) \left((1 + r_R) \right). \end{aligned} \quad (28)$$

Then we can get the optimal solutions of the retailers, the manufacturer and the remanufacturer.

Theorem 7. When the remanufacturer received a loan from the bank and received subsidies from government, the optimal retail prices, the optimal production quantities (market demands) of the new products and the reman, the optimal wholesale price, the optimal profits of manufacturer and remanufacturer are given in the online supplements.

Proof. See Online Supplements.

Retailer financing (Scenario 112)

Same as 3.4.1, if the remanufacturer and consumers receive subsidies from the government, we can get the total market demand of the new product and the reman.

$$q_r^b = \lambda \left(\frac{p_n - p_r + \tau - v + \rho p_r}{1 - \gamma} - \frac{p_r - \tau - \rho p_r}{\gamma} \right)$$

$$q_r^b = \lambda \left(\frac{p_n - p_r + \tau - v + \rho p_r}{1 - \gamma} - \frac{p_r - \tau - \rho p_r}{\gamma} \right)$$

$$q_n = q_n^a + q_n^b = 1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}$$

$$q_n = q_n^a + q_n^b = 1 - \frac{p_n - p_r + \rho p_r + \lambda(\tau - v)}{1 - \gamma}$$

Since the remanufacturer can receive a loan from the retailer, the two retailers' profit functions can be written as follow:

$$\max \Pi_{R1}^{112}(p_n) = q_n(p_n - w_n)$$

$$\max \Pi_{R2}^{112}(p_r) = q_r(p_r - w_r) + (q_r c_r - B) \left((1 + r_R) \right)$$

Because the manufacturer and remanufacturer are competitive and the remanufacturer can receive government subsidies, the profit functions of the manufacturer and the remanufacturer can be written as:

$$\max \Pi_M^{112}(w_n, w_r) = q_n(w_n^{112} - c_n)$$

$$\begin{aligned} & \max \Pi_{RM}^{112}(w_n, w_r) \\ & = q_r^{112} \left(w_r^{112} - c_r + s \right) + q_r^{112} \left(w_r^{112} - c_r - \tau + s \right) - \left(c_r q_r^{112} - B \right) \\ & \quad \left(1 + r_R \right) \end{aligned}$$

Theorem 8. When customers receive a rebate and remanufacturers receive a remanufacturing subsidy, the optimal retail prices, the optimal production quantities (market demands) of the new products and the reman, the optimal wholesale price, the optimal profits of manufacturer and remanufacturer are given in the online supplements.

Proof. Please see the Online Supplements.

In the next section, we conduct numerical experiments to draw further insights from above, and examine the influences of ρ , γ and s on the optimal strategies and SC performance.

Numerical studies

In Section 3, we considered eight scenarios based on whether there exists manufacturer-remanufacturer competition, government incentives and capital constraints. We derived analytical relationships between input parameters (i.e., customer rebate ratio and reman subsidy/unit) and various performance metrics. The SC parameters could significantly impact the optimal profits, product quantities, and prices, which in turn affect the success of recycling programs. Inadequate parameter settings worsen SC performance, and subsequently place the success of the remanufacturing initiatives at risk. We now study the impact of input parameter changes on retail prices, product demand and SC performance. In order to study and understand it easily, we divide all scenarios into two groups: no financial constraints and financial constraints.

Effect of customer rebate ratio (ρ) on retail prices

To numerically investigate the impact of changes in ρ on the retail prices, we set $\gamma = 0.9$, $c_n = 0.3$, $c_r = 0.2$, $\lambda = 0.5$, $\tau = 0.08$, $v = 0.05$, $s = 0.01$, $r_B = 0.1$ and $r_R = 0.3$.

Fig. 2 shows the cases of without government-incentive and government-incentive (scenarios 010 and 110) for comparison. In Scenario 110 (with subsidy/rebate, competition), when rebate ratio (ρ) increases, the retail price of the reman increases rapidly, while that of the new product decreases slowly and marginally. The decrease in the new product's retail price is due to more competition from higher customer's reman rebate.

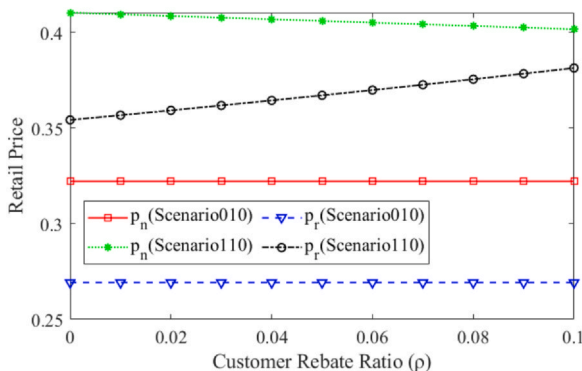


Fig. 2. Effect of ρ on retail prices.

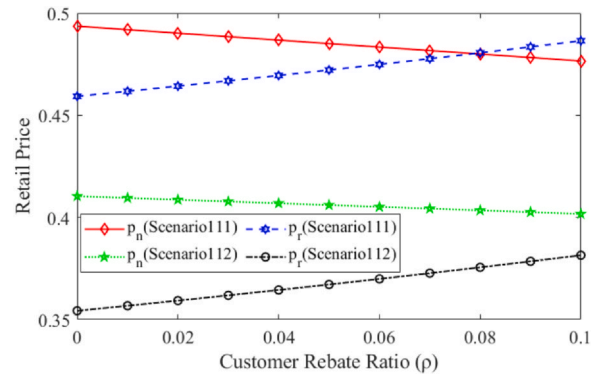


Fig. 3. Effect of ρ on retail prices.

Same as before, we set $\gamma = 0.9$, $c_n = 0.3$, $c_r = 0.2$, $\lambda = 0.5$, $\tau = 0.08$, $v = 0.05$, $s = 0.01$, $r_B = 0.1$ and $r_R = 0.3$, and we can see the impact of customer rebate ratio change on retail price when there are capital constraints. The results are similar to Fig. 2. As Fig. 3 shows, when rebate ratio (ρ) increases, the retail price of the reman increases, while the retail price of the new products decreases slowly. And we can find that the retail prices of the reman and the new products under BF scenario are higher than that under RF scenario. This relates to the intensified competition between the two kinds of products and the different financing strategies' interest rate.

Effect of Customer Rebate Ratio (ρ) on Demand

Figs. 4 and 5 show the effect of changes in ρ on demand under four scenarios (scenarios 100, 110, 111 and 112). The Fig. 4 shows that when there are government incentives (scenarios 100 and 110), ρ has a positive impact on reman demand, but has an adverse impact on the new product demand. The Fig. 5 shows that, for the two scenarios (scenarios 111 and 112), the customer rebate ratio has a positive influence on the reman demand, while it has a negative impact on the new demand, which is consistent with the previous conclusion.

Fig. 6 shows that the total demand increases with the customer rebate in Scenarios 110, 111 and 112. Fig. 6 also shows that the total demand is little affected by consumer rebate in Scenario 100 (i.e., the no competition case with government incentives). According to more accurate data, the total demand has only a slight increase in

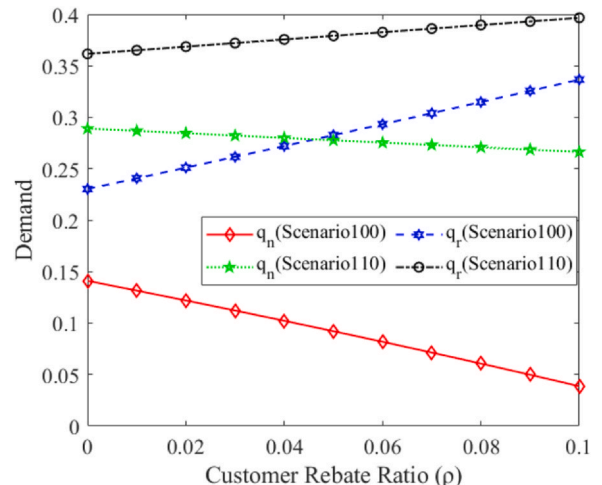


Fig. 4. Effect of ρ on product demand (Scenario100&110).

Scenario 100. Fig. 6 also shows that the total demand under competition is larger than that under cooperation, which means that the competition between the manufacturer and the remanufacturer can induce the customers to buy more products. And we can also find that, the total demand under RF scenario is larger than that under BF scenario, which means the RF strategy is better for the SC. By comparing Fig. 4, Fig. 5 and Fig. 6, we can find that the rebate policy will cannibalize new product demand, but it has a positive impact on the total sales volume of new products and remanufactured products.

Effect of Remanufacturing Subsidy (s) on Retail Prices, Demand and SC Performance

Assuming $\rho = 0.1$ and $B = 1$ along with the parameters defined above, we investigate the impact of remanufacturing subsidy (s) on the optimal retail prices, the demands, and the profits of the manufacturer and the remanufacturer. Fig. 7 shows that the retail prices of new product and reman will always decrease with s . This is because a larger s induces the remanufacturer to lower remans' retail price, which in turn puts a downward pressure on the new product's retail price. Observe that the retail price of the reman decreases at a faster rate than that of the new product. As expected, the retail prices are higher in the no competition scenarios than those in the competition ones.

Fig. 8 shows that under financing situation, the retail prices for both new products and the reman will always decrease with

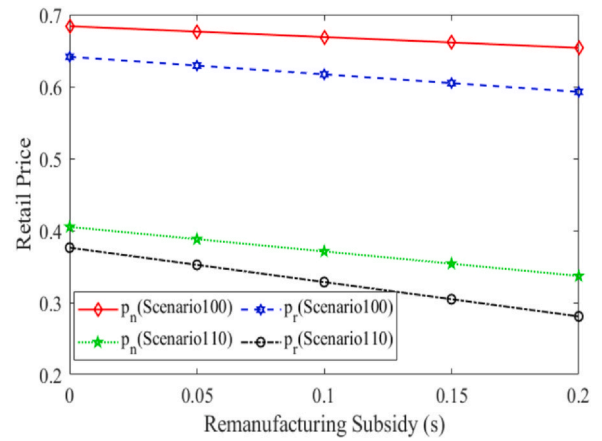


Fig. 7. Effect of s on retail prices (Scenario 100 & 110).

remanufacturing subsidy, whether the BF scenario or the RF scenario. This is because when the remans' retail price decreases with the remanufacturing subsidy, it also puts a downward pressure on the new products' retail price, which is similar as before. We can also observe that the products' retail price under BF scenario is higher than that under RF scenario, this is because the total demand for products under BF scenario is smaller than that under RF scenario. So if the SC members under BF scenario want to make more profits, they need to set a higher retail price.

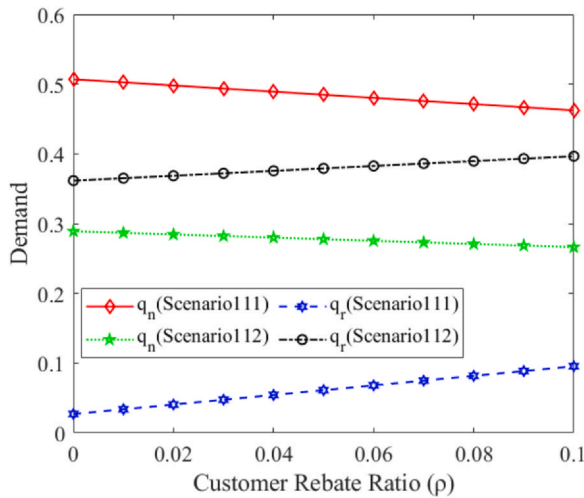


Fig. 5. Effect of ρ on product demand (Scenario 111 & 112).

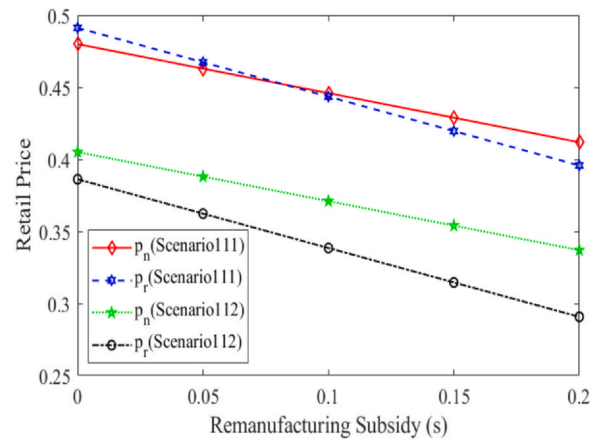


Fig. 8. Effect of s on retail prices (Scenario 111 & 112).

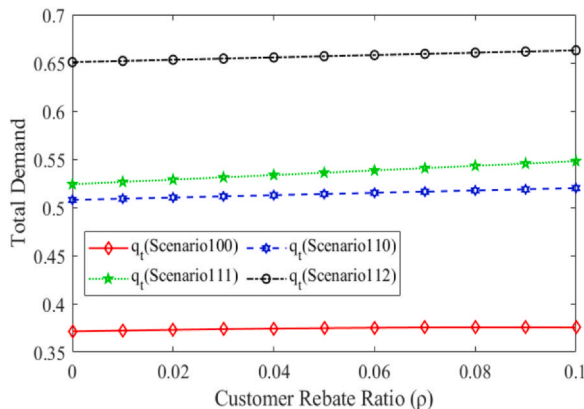


Fig. 6. Effect of ρ on total demand.

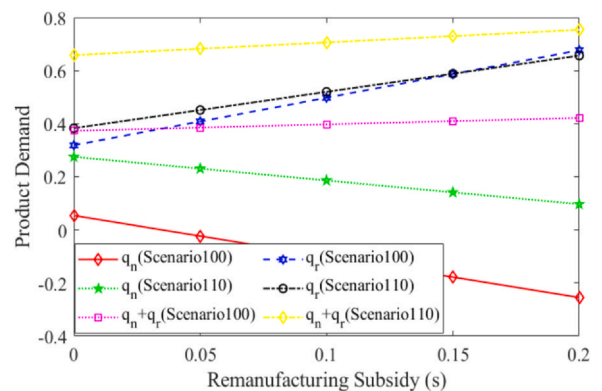


Fig. 9. Effect of s on demand.

Fig. 9 shows that the demands increase with s for new products, regardless of competition. We can also observe that the total demand q_i increases with s . Note that the demand under no competition situation is smaller than that under competition situation. This is because the retail prices under no competition situation are higher than those under competition situation. Fig. 10 shows that the total profits of the manufacturer and the remanufacturer increase with s ; also, if there are no capital constraints, no competition leads to higher profits than the competition case. We can also observe that the total profits of the manufacturer and the remanufacturer under BF scenario are larger than those under RF scenario. This is because although the total demand of products under BF scenario is smaller, the retail prices of new products and the reman under BF scenario are much higher than those under RF scenario.

Comparison of Customer Rebate and Remanufacturing Subsidy

We now study the effect of customer rebate ratio (ρ) on the profits of the manufacturer, the remanufacturer and total profits for specific s values under Scenario 100 and Scenario 110. We set $\gamma = 0.9$, $c_n = 0.3$, $c_r = 0.2$, $\lambda = 0.5$, $\tau = 0.08$ and $v = 0.05$. Fig. 11 shows that, the higher government subsidies lead to higher total profits. Fig. 11 also show that both the customer rebate and remanufacturing subsidy improve the total profit of remanufacturer and manufacturer. Next, we analyze the situation when the remanufacturer takes financing decisions. Fig. 12 shows that no matter whether the remanufacturer adopts BF scenario or RF scenario, the manufacturer's profit always decreases with the customer rebate ratio. Fig. 12 also shows that the profit of the manufacturer under BF scenario is higher than that under MF scenario. Fig. 13 shows that with the same government subsidies, the profit of the remanufacturer under RF scenario is

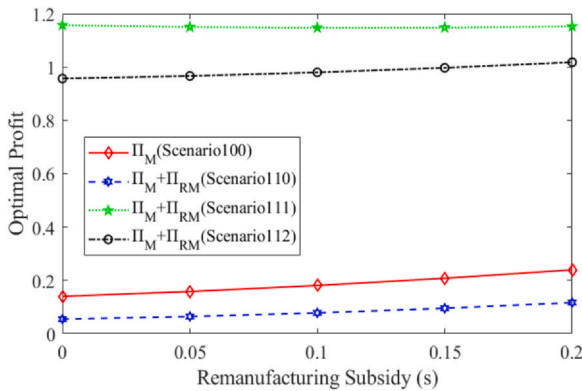


Fig. 10. Effect of s on profit.

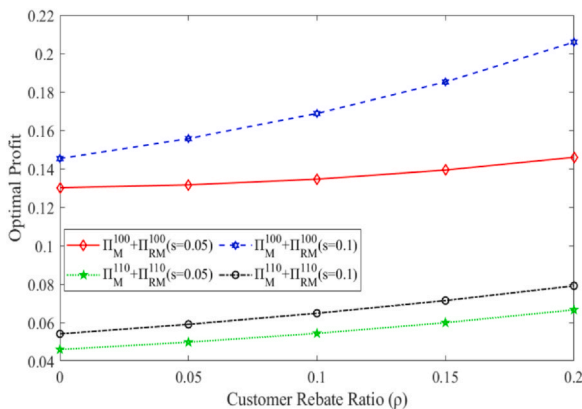


Fig. 11. Effect of ρ on the total profits.

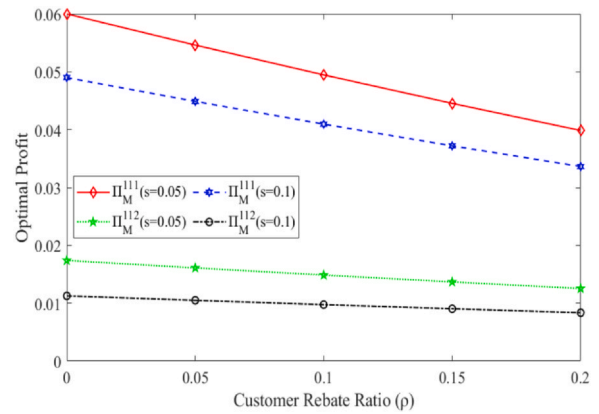


Fig. 12. Effect of ρ on profits of the manufacturer.

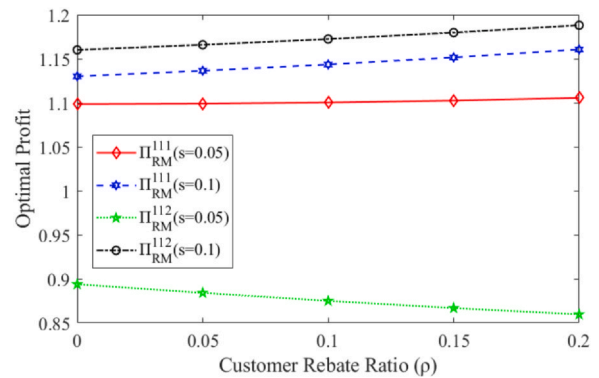


Fig. 13. Effect of ρ on profits of the remanufacturer.

higher than that under BF scenario. In addition, when the remanufacturer takes the BF scenario, the profit of the remanufacturer increases with the rebate ratio. And for RF scenario, we can observe that when $s = 0.1$, the profit of remanufacturer increases with ρ , whereas when $s = 0.05$, the profit of remanufacturer decreases with ρ . These results can help policy makers decide how to institute effective government incentive programs.

Conclusions

Remanufacturing activity continues to grow globally, e.g., the current European remanufacturing is near \$30 billion, and is expected to reach 90 billion by 2030 [8]. Consequently, remanufacturing has drawn considerable attention for policy-making in practice and in research. Governments are initiating incentives to encourage remanufacturing and increase end-customer demand for the remanufactured products. However, few researchers have noticed that many downstream manufacturing enterprises are SMEs, which are facing the problem of capital shortage. To our knowledge, this study is the first to study the impact of remanufacturing financing and incentives (customer rebate and remanufacturing subsidy) on the remanufacturing SC with the TOR program.

In this paper, we use a competitive model to address the optimal financing and pricing strategy when the government implements remanufacturing subsidies. The ensuing numerical experiments study the effect of the customer rebate ratio on the retail prices and the demands of the products. Then, we investigate the influences of remanufacturing subsidy on the retail prices, the quantities demanded, and SC performance. We numerically show how the remanufacturing subsidy and the customer rebate ratio impact remanufacturing activity, and government incentives are conducive

to the expansion of SC. The next analysis shows that RF strategy is more favorable for the remanufacturer, while the BF strategy is better for the whole SC. Our research provides managerial insights into the impacts of different subsidy policies and provides recommendations for the government on how to provide policy incentives. In addition, when remanufacturers in the SC encounter financial difficulties, the most suitable financing strategy can also be selected based on the research results of this study.

This paper innovatively studies the government's subsidy policies and financing decisions of remanufacturers with the TOR program, providing management advices to the government and firms for sustainable development. There are still many limitations of this study, such as only considering two financing strategies, and in fact, remanufacturers may have more financing options. For example, the supply chain members can choose mixed financing methods to solve their funding problems. Furthermore, we have not considered the bankruptcy risk of the remanufacturer.

Future research could examine the impacts of reman quality on customer heterogeneity. We can also look into the potential sharing of the remanufacturing subsidy between the remanufacturer and the

retailer. Another direction of research is to investigate from a policy maker's perspective where the total budget available for the remanufacturing subsidy and the customer rebate is limited. The decision then is to determine the best allocation of the budget that will maximize the remanufacturing activities.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Technical Appendix for: financing strategies and government incentives in a competing supply chain with trading-old-for-remanufactured programs

Proof of Theorem 1.

Since $\frac{\partial^2 \Pi_{R1}^{000}(p_n)}{\partial p_n^2} = -\frac{2}{1-\gamma} < 0$, $\frac{\partial^2 \Pi_{R2}^{000}(p_r)}{\partial p_r^2} = -\frac{2}{1-\gamma} - \frac{2}{\gamma} < 0$, Π_{R1}^{000} and Π_{R2}^{000} are strictly concave in p_n and p_r respectively. We can identify the optimal product prices by solving the first-order conditions of $\Pi_{R1}(p_n)$ and $\Pi_{R2}(p_r)$:

$$p_r^{000} = \frac{\lambda\tau\gamma + \gamma\lambda v - \gamma w_n^{000} - 2\lambda\tau - 2w_r^{000}}{\gamma - 4} \quad (A1)$$

$$p_n^{000} = \frac{\gamma\lambda v - \lambda\tau + 2\lambda v - 2w_n^{000} - w_r^{000}}{\gamma - 4} \quad (A2)$$

Substituting the optimal prices from Eqs. (A1) and (A2) into the demand functions of q_n and q_r , we have

$$q_n^{000} = \frac{\gamma^2 + \left(-\lambda v + w_n^{000} - 5\right)\gamma + \left(-\tau + 2v\right)\lambda - 2w_n^{000} + w_r^{000} + 4}{(\gamma - 4)(-1 + \gamma)} \quad (A3)$$

$$q_r^{a000} = \frac{-\lambda\left(\lambda\tau + v - w_n^{000}\right)\gamma^2 + \left(\left(3\tau - 2v\right)\lambda^2 + \left(3v - 2w_n^{000} + w_r^{000}\right)\lambda + w_n^{000} + w_r^{000}\right)\gamma - 2\lambda\tau - 2w_r^{000}}{(\gamma - 4)(-1 + \gamma)\gamma} \quad (A4)$$

$$q_r^{b000} = -\frac{\left(\left(\lambda v - 2\tau + v\right)\gamma^2 + \left(-3\lambda v + 9\tau - 4v - w_n^{000} - w_r^{000}\right)\gamma + 2\lambda\tau - 4\tau + 2w_r^{000}\right)\lambda}{(\gamma - 4)(-1 + \gamma)\gamma} \quad (A5)$$

From the above results, it is easy to check the impact of the wholesale prices on the product demands. As w_r increases, the optimal new product demand increases but the reman demand decreases. Similarly, as w_n increases, the optimal reman demand increases but the new product demand decreases.

Let us now determine the optimal wholesale prices. Given the retailers' reaction functions (Eqs. (A1)–(A2)), the manufacturer's problem will be changed as:

$$\max_{w_r, w_n} \Pi_M^{000}(w_n, w_r) = q_r^{a000}(w_r - c_r) + q_r^{b000}(w_r - c_r - \tau) + q_n^{000}(w_n - c_n) \quad (A6)$$

Because $0 < \gamma < 1$ and $0 < \lambda < 1$, we have $\frac{\partial^2 \Pi_M^{000}}{\partial w_n^2} = \frac{2\gamma(\gamma-2)}{(\gamma-4)(\gamma-1)\gamma} < 0$, $\frac{\partial^2 \Pi_M^{000}}{\partial w_r^2} = \frac{4\lambda(\gamma-1) + 2(\gamma-2)}{(\gamma-4)(\gamma-1)\gamma} < 0$ and $\frac{\partial^2 \Pi_M^{000}}{\partial w_n^2} \times \frac{\partial^2 \Pi_M^{000}}{\partial w_r^2} - \frac{\partial^2 \Pi_M^{000}}{\partial w_n \partial w_r}^2 = \frac{-\lambda^2\gamma^2 + (\lambda+2)^2\gamma - 16\lambda - 16}{\gamma(\gamma-1)(\gamma-4)^2} > 0$. This suggests that Π_M^{000} is concave in w_n and w_r , so we can obtain the optimal wholesale prices of the remanufactured and new products by solving

$$\frac{\partial \Pi_M^{000}(w_n, w_r)}{\partial w_r} = \frac{\left((-\tau - v)\lambda^2 + (2\tau - 2v + w_n)\lambda \right) \gamma^2 + \left((3\tau + v)\lambda^2 + (4w_r - 10\tau + 7v - 2c_r - w_n)\lambda \right) \gamma}{(\gamma - 4)(\gamma - 1)\gamma} + \frac{-2\lambda^2\tau + (-4w_r + 4\tau + 2c_r)\lambda - w_r + 2c_r + (2w_r - c_n - c_r + 2w_n)\gamma}{(\gamma - 4)(\gamma - 1)\gamma} = 0 \quad (A7)$$

$$\frac{\partial \Pi_M^{000}(w_n, w_r)}{\partial w_n} = \frac{\gamma^3 + \left((w_r - c_r - v)\lambda + 2w_n - c_n - 5 \right) \gamma^2 + (-w_r - 2\tau + c_r + 2v)\lambda \gamma}{(\gamma - 4)(\gamma - 1)\gamma} + \frac{(2w_r - c_r - 4w_n + 2c_n + 4)\gamma}{(\gamma - 4)(\gamma - 1)\gamma} = 0 \quad (A8)$$

Solving Eqs. (A7) and (A8), we find that the optimal wholesale prices are

$$w_n^{000} = \frac{1}{(-1 + \gamma) \left((\gamma^2 - \gamma)\lambda^2 + (16 - 4\gamma)\lambda - 4\gamma + 16 \right)} \left((-1 + \gamma)^2 \left(\tau + v \right) \gamma - 2\tau \right) \lambda^3 - 2 \left(\gamma - 1 \right) \left(\tau - v \right) \gamma^2 + \left(-6\tau + 4.5v + c_r \right) \gamma + 8\tau - 4v - c_r \lambda^2 + \left(4\gamma^3 + (-4\tau + 2v - 3c_n - c_r - 24) \right) \gamma^2 + (16\tau - 6v + 11c_n + 3c_r + 36)\gamma - 8v - 8c_n - 2c_r - 16 \lambda + 2(\gamma - 4)(-1 + \gamma)(\gamma - c_n - 2)$$

$$w_r^{000} = \frac{1}{(-1 + \gamma) \left(\gamma^2 \lambda^2 - (\lambda + 2)^2 \gamma + 16\lambda + 16 \right)} \left(-\gamma^4 \lambda + \left(-2 + (-2\tau - v + c_r) \right) \lambda^2 + \left(4\tau - 4v + c_n + 6 \right) \lambda \right) \gamma^3 + ((12\tau + 3v - 2c_r)\lambda^2 + (-28\tau + 24v - 3c_n - c_r - 9)\lambda - 2c_r + 10)\gamma^2 + ((-18\tau - 2v + c_r)\lambda^2 + (52\tau - 32v + 2c_n + 9c_r + 4)\lambda + 10c_r - 8)\gamma + 8\lambda^2\tau + (-16\tau - 8c_r)\lambda - 8c_r$$

Substituting the optimal wholesale prices from above into the Eqs. (A1)–(A5), we have the optimal retail price and the optimal product demand. Then we can substitute these formulas into the manufacturer's profit function and we have

$$\max_{w_r, w_n} \Pi_M^{000} = \frac{-\lambda \left(\lambda\tau + v - w_n^{000} \right) \gamma^2 + \left((3\tau - 2v)\lambda^2 + (3v - 2w_n^{000} + w_r^{000})\lambda + w_n^{000} + w_r^{000} \right) \gamma - 2\lambda\tau - 2w_r^{000}}{(\gamma - 4)(-1 + \gamma)\gamma} \left(w_r^{000} - c_r \right) + \frac{\left((\lambda v - 2\tau + v)\gamma^2 + (-3\lambda v + 9\tau - 4v - w_n^{000} - w_r^{000})\gamma + 2\lambda\tau - 4\tau + 2w_r^{000} \right) \lambda}{(\gamma - 4)(-1 + \gamma)\gamma} \left(w_r^{000} - c_r - \tau \right) + \frac{\gamma^2 + (-\lambda v + w_n^{000} - 5)\gamma + (-\tau + 2v)\lambda - 2w_n^{000} + w_r^{000} + 4}{(\gamma - 4)(-1 + \gamma)} \left(w_n^{000} - c_n \right).$$

Proof of Theorem 2.

To determine the optimal wholesale prices, we take the first derivative of $\Pi_{RM}^{010}(w_r)$ and $\Pi_M^{010}(w_n)$ with respect to w_r and w_n , then set them to zero:

$$\frac{\partial \Pi_{RM}^{010}(w_r)}{\partial w_r} = \frac{\left((-\tau - v)\gamma^2 + (3\tau + v)\gamma - 2\tau \right) \lambda^2 + \left((2\tau - 2v + w_n)\gamma^2 + (4w_r - 10\tau + 7v - 2c_r - w_n)\lambda \right) \gamma \lambda}{(\gamma - 4)(\gamma - 1)\gamma} - \frac{(\gamma - 2)(-w_r + c_r) + (w_n + w_r)\gamma + 2w_r - (4w_r + 4\tau + 2c_r)\gamma}{(\gamma - 4)(\gamma - 1)\gamma} = 0 \quad (A9)$$

and

$$\frac{\partial \Pi_M^{010}(w_n)}{\partial w_n} = \frac{\gamma^2 + (-\lambda v + w_n - 5)\gamma + (2v - \tau)\lambda - 2w_n + w_r + 4 - (\gamma - 2)(c_n - w_n)}{(\gamma - 4)(\gamma - 1)} = 0 \quad (A10)$$

Because $0 < \gamma < 1$ and $0 < \lambda < 1$, we have $\frac{\partial^2 \Pi_M^{010}}{\partial w_n^2} = \frac{2(\gamma - 2)}{(\gamma - 4)(\gamma - 1)} < 0$ and $\frac{\partial^2 \Pi_{RM}^{010}}{\partial w_r^2} = \frac{4(\gamma - 1)\lambda + 2(\gamma - 2)}{(\gamma - 4)(\gamma - 1)\gamma} < 0$. This suggests that Π_M^{010} is concave in w_n and Π_{RM}^{010} is concave in w_r . Combining Eqs. (A9) and (A10), we can find the equilibrium wholesale prices:

$$w_n^{010} = \frac{1}{\left(\frac{7\lambda+4}{\gamma^2} + \frac{-23\lambda-17}{\gamma+16\lambda+16}\right)} \left(\left(-4\lambda-2 \right) \gamma^3 + \left(\left(-\tau+3v \right) \lambda^2 + \left(2\tau+4c_n+24 \right) \lambda + 2c_n+14 \right) \gamma^2 \right. \\ \left. + ((7\tau-11v)\lambda^2 + (-8\tau-v-12c_n-2c_r-36)\lambda - 8c_n-c_r-28)\gamma + (-6\tau+8v)\lambda^2 + (8v+8c_n+2c_r+16)\lambda \right. \\ \left. + 8c_n+2c_r+16 \right) \\ w_r^{010} = \frac{1}{\left(\frac{7\lambda+4}{\gamma^2} + \frac{-23\lambda-17}{\gamma+16\lambda+16}\right)} \left(\gamma^4\lambda + \left(1 + \left(2\tau+v \right) \lambda^2 + \left(4v-4\tau-c_n-6 \right) \lambda \right) \gamma^3 + \left(\left(-11\tau-3v \right) \lambda^2 \right. \right. \\ \left. \left. + (28\tau-23v+3c_n+4c_r+9)\lambda - c_n+2c_r-5 \right) \gamma^2 + ((17\tau+2v)\lambda^2 + 30v-2c_n-12c_r-4-49\tau)\lambda + 2c_n-8c_r+4 \right) \gamma \\ - 8\lambda^2\tau + (16\tau+8c_r)\lambda + 8c_r$$

Substituting the optimal wholesale prices from above into the Eqs. (A1)–(A5), we have the optimal retail price and the optimal product demand. Then we can obtain the manufacturer's optimal profit and the remanufacturer's optimal profit:

$$\max_{w_r, w_n} \Pi_M^{010} = \frac{\left(-w_n^{010} + c_n \right) \left(\gamma^2 + \left(-\lambda v + w_n^{010} - 5 \right) \gamma + \left(-\tau + 2v \right) \right) - 2w_n^{010} + w_r^{010} + 4}{(\gamma-4)(\gamma-1)}, \\ \max_{w_r, w_n} \Pi_{RM}^{010} = \frac{\left(-w_r^{010} + c_r \right) \left(\lambda - 1 \right) \left(\left(-\gamma^2 + \gamma \right) \lambda \left(\tau - v \right) + \lambda \left(\tau - 2v \right) \gamma^2 + \left(\left(-\tau + 4v \right) \lambda + w_n^{010} + w_r^{010} \right) \gamma - 2\lambda\tau - 2w_r^{010} \right)}{(\gamma-4)(\gamma-1)\gamma}.$$

Proof of Theorem 3.

Since $\frac{\partial^2 \Pi_{R1}^{100}(p_n)}{\partial p_n^2} = -\frac{2}{1-\gamma} < 0$ and $\frac{\partial^2 \Pi_{R2}^{100}}{\partial p_r^2} = 2(\rho-1) \left(\frac{1}{1-\gamma} + \frac{1}{\gamma} \right) < 0$, Π_{R1}^{100} and Π_{R2}^{100} are strictly concave in p_n and p_r respectively. We can identify the optimal product prices by solving the first-order conditions of $\Pi_{R1}(p_n)$ and $\Pi_{R2}(p_r)$:

$$p_n^{100} = \frac{\gamma\lambda v + \rho w_r^{100} + \lambda\tau - 2\lambda v + 2\gamma - 2w_n^{100} - w_r^{100} - 2}{\gamma-4} \quad (A11)$$

$$p_r^{100} = -\frac{\gamma\lambda\tau + \gamma\lambda v + \gamma^2 - \gamma w_n^{100} + 2\rho w_r^{100} - 2\lambda\tau - \gamma - 2w_r^{100}}{\gamma\rho - \gamma - 4\rho + 4} \quad (A12)$$

Substituting the optimal price values from Eqs. (A11) and (A12) into the demand functions of q_n and q_r , we have:

$$q_n^{100} = 1 - \frac{\gamma^2 + \left(\lambda v - w_n^{100} - 3 \right) \gamma + \left(\tau - 2v \right) \lambda + 2w_n^{100} + 2 + \left(-1 + \rho \right) w_r^{100}}{(-1+\gamma)(\gamma-4)} \quad (A13)$$

$$q_r^{a100} = \frac{\left(-1 + \lambda \right) \left(\left(\lambda v + 1 \right) \gamma^2 + \left(\left(-1 + \rho \right) w_r^{100} - 3\lambda v - w_n^{100} - 1 \right) \gamma + \left(-2\rho + 2 \right) w_r^{100} + 2\lambda\tau \right)}{\gamma(-1+\gamma)(\gamma-4)} \quad (A14)$$

$$q_r^{b100} = -\frac{\lambda \left(\left(\lambda v - v + 1 \right) \gamma^2 + \left(\left(-1 + \rho \right) w_r^{100} - 3\lambda v + \tau + 4v - w_n^{100} - 1 \right) \gamma + \left(-2\rho + 2 \right) w_r^{100} + 2\tau \left(\lambda - 2 \right) \right)}{\gamma(-1+\gamma)(\gamma-4)} \quad (A15)$$

The manufacturer's problem, with the objective of maximizing the total profit from selling the new and remanufactured products, can be expressed as

$$\max_{w_r, w_n} \Pi_M^{100} = q_r^{a100} \left(w_r - c_r + s \right) + q_r^{b100} \left(w_r - c_r - \tau + s \right) + q_n^{100} \left(w_n - c_n \right). \quad (A16)$$

Since $\frac{\partial^2 \Pi_M^{100}}{\partial w_r^2} = \frac{2(\rho-1)(2-\gamma)}{(\gamma-4)(\gamma-1)\gamma} < 0$, $\frac{\partial^2 \Pi_M^{100}}{\partial w_n^2} = \frac{2\gamma(\gamma-2)}{(\gamma-4)(\gamma-1)\gamma} < 0$ and $\frac{\partial^2 \Pi_M^{100}}{\partial w_r^2} * \frac{\partial^2 \Pi_M^{100}}{\partial w_n^2} - \frac{\partial^2 \Pi_M^{100}}{\partial w_n \partial w_r} = \frac{4(1-\rho)\gamma^2 - (-\rho^2 + 20\rho - 20)\gamma - 16\rho + 16}{\gamma(\gamma-4)^2(\gamma-1)^2} > 0$, Π_M^{100} are strictly concave in w_r and w_n respectively. To obtain the optimal wholesale prices, we take the first derivative of Π_M^{100} with respect to w_r and w_n , and set them to equal zero:

$$\frac{\partial \Pi_M^{100}}{\partial w_r} = \frac{-\gamma^2 + \left(\left((\rho - 2)\tau - v \right) \lambda + \left(2 - 2\rho \right) w_r + \left(2 - \rho \right) w_n + \left(\rho - 1 \right) c_r + \left(c_n - s \right) \rho + s - c_n + 1 \right) \gamma}{(\gamma - 4)(\gamma - 1)\gamma} + \frac{(4 - 2\rho)\tau\lambda + (2\rho - 2)(w_r - c_r + s) + (2\rho - 2)w_r}{(\gamma - 4)(\gamma - 1)\gamma} = 0 \quad (\text{A17})$$

$$\frac{\partial \Pi_M^{100}}{\partial w_n} = \frac{(-\lambda v - c_n + 2w_n - 2)\gamma^2 + ((-2\tau + 2v)\lambda + (2 - \rho)w_r - 4w_n + s + 2c_n - c_r + 2)\gamma}{(\gamma - 4)(\gamma - 1)\gamma} = 0 \quad (\text{A18})$$

Solving Eqs. (A17) and (A18), we obtain optimal wholesale prices of new and remanufactured products, i.e., w_r^{100*} and w_n^{100*} .

$$w_n^{100} = \frac{1}{\gamma\rho^2 + (4\gamma^2 - 20\gamma + 16)\rho - 4\gamma^2 + 20\gamma - 16} \left(\left(\left(\lambda\tau + c_n + c_r - s \right) \gamma - 2\lambda\tau + 2s - 2c_r \right) \rho^2 + \left(\left(2\lambda v + 2c_n + 3 \right) \gamma^2 + (-9\lambda v - 11c_n - c_r + s - 11)\gamma + 8\lambda v - 2s + 8c_n + 2c_r + 8 \right) \rho - 2(\gamma - 1)(\gamma - 4)(\lambda v + c_n + 1) \right)$$

$$w_r^{100} = \frac{1}{(4\rho - 4)\gamma^2 + (\rho^2 - 20\rho + 20)\gamma + 16\rho - 16} \left(-2\gamma^3 + \left(\left((2\tau - v)\lambda - 2s + c_n + 2c_r - 2 \right) \rho - 4\lambda\tau + 2s - 2c_r + 10 \right) \gamma^2 + (((-10\tau + 2v)\lambda + 9s - 2c_n - 9c_r + 2)\rho + 20\lambda\tau - 10s + 10c_r - 8)\gamma + (8\lambda\tau + 8c_r - 8s)\rho - 16\lambda\tau + 8s - 8c_r \right)$$

Substituting the optimal wholesale prices from above into the Eq. (A11)–(A15), we have the optimal retail price and the optimal product demand. Then we can obtain the optimal total profit of manufacturer and remanufacturer:

$$\max \Pi_M^{100} = \frac{(-1 + \lambda) \left((\lambda v + 1)\gamma^2 + ((-1 + \rho)w_r^{100} - 3\lambda v - w_n^{100} - 1)\gamma + (-2\rho + 2)w_r^{100} + 2\lambda\tau \right) (w_r^{100} - c_r + s)}{\gamma(-1 + \gamma)(\gamma - 4)} - \frac{\lambda \left((\lambda v - v + 1)\gamma^2 + ((-1 + \rho)w_r^{100} - 3\lambda v + \tau + 4v - w_n^{100} - 1)\gamma + (-2\rho + 2)w_r^{100} + 2\tau(\lambda - 2) \right) (w_r^{100} - c_r - \tau + s)}{\gamma(-1 + \gamma)(\gamma - 4)} + \left(1 - \frac{\gamma^2 + (\lambda v - w_n^{100} - 3)\gamma + (\tau - 2v)\lambda + 2w_n^{100} + 2 + (-1 + \rho)w_r^{100}}{(-1 + \gamma)(\gamma - 4)} \right) (w_n^{100} - c_n).$$

Then we can derive the Theorem 3.

Proof of Theorem 4.

Same as Theorem 3, we can get the optimal retail price and quantities.

$$q_n^{110} = 1 - \frac{\gamma^2 + (\lambda v - w_n^{110} - 3)\gamma + (\tau - 2v)\lambda + 2w_n^{110} + 2 + (-1 + \rho)w_r^{110}}{(-1 + \gamma)(\gamma - 4)}$$

$$q_r^{a110} = \frac{(-1 + \lambda)((\lambda v + 1)\gamma^2 + ((-1 + \rho)w_r^{110} - 3\lambda v - w_n^{110} - 1)\gamma + (-2\rho + 2)w_r^{110} + 2\lambda\tau)}{\gamma(-1 + \gamma)(\gamma - 4)}$$

$$q_r^{b110} = -\frac{\lambda((\lambda v - v + 1)\gamma^2 + ((-1 + \rho)w_r^{110} - 3\lambda v + \tau + 4v - w_n^{110} - 1)\gamma + (-2\rho + 2)w_r^{110} + 2\tau(\lambda - 2))}{\gamma(-1 + \gamma)(\gamma - 4)}, p_n^{110} = \frac{\gamma\lambda v + \rho w_r^{110} + \lambda\tau - 2\lambda v + 2\gamma - 2w_n^{110} - w_r^{110} - 2}{\gamma - 4}$$

$$p_r^{110} = -\frac{\gamma\lambda\tau + \gamma\lambda v + \gamma^2 - \gamma w_n^{110} + 2\rho w_r^{110} - 2\lambda\tau - \gamma - 2w_r^{110}}{\gamma\rho - \gamma - 4\rho + 4}$$

The profits of manufacturer and remanufacturer can be written as the following two formulas respectively:

$$\max \Pi_M^{110} = q_n^{110} \left(w_n - c_n \right) = \left(1 - \frac{\gamma^2 + (\lambda v - w_n^{110} - 3)\gamma + (\tau - 2v)\lambda + 2w_n^{110} + 2 + (-1 + \rho)w_r^{110}}{(-1 + \gamma)(\gamma - 4)} \right) (w_n - c_n),$$

$$\max \Pi_{RM}^{110} = q_r^{a110} (w_r - c_r + s) + q_r^{b110} (w_r - c_r - \tau + s)$$

To determine the optimal wholesale prices, we take the first derivative of $\Pi_{RM}^{110}(w_r)$ and $\Pi_M^{110}(w_n)$ with respect to w_r and w_n , then set them to zero to obtain

$$\frac{\partial \Pi_M^{110}}{\partial w_n} = -\frac{(\lambda v - w_n + 2)\gamma + 2w_n + (\tau - v)\lambda - 2 + (\rho - 1)w_r + (c_n - w_n)(2 - \gamma)}{(\gamma - 1)(\gamma - 4)} = 0 \quad (\text{A19})$$

$$\begin{aligned} \frac{\partial \Pi_{RM}^{110}}{\partial w_r} = & \frac{-\gamma^2 + \left(\left((\rho - 2)\tau - v \right) \lambda - (\rho - 1) \left(w_r - c_r + s \right) - (\rho - 1) w_r + w_n + 1 \right) \gamma - 2(\rho - 2) \tau \lambda}{(\gamma - 4)(\gamma - 1)\gamma} \\ & + \frac{2(\rho - 1)(w_r - c_r + s) + 2(\rho - 1)w_r}{(\gamma - 4)(\gamma - 1)\gamma} = 0 \end{aligned} \quad (\text{A20})$$

Since $\frac{\partial^2 \Pi_M^{110}}{\partial w_n^2} = -\frac{2(2-\gamma)}{(\gamma-1)(\gamma-4)} < 0$ and $\frac{\partial^2 \Pi_{RM}^{110}}{\partial w_r^2} = \frac{2(\rho-1)(2-\gamma)}{(\gamma-4)(\gamma-1)\gamma} < 0$, Π_M^{110} and Π_{RM}^{110} are strictly concave in w_r and w_n respectively. Combining Eqs. (A19) and (A20), we find

$$\begin{aligned} w_n^{110} = & \frac{2\gamma^2\lambda v + \gamma\rho\lambda\tau + 2\gamma^2c_n + \gamma\rho c_r - \gamma\rho s - 9\gamma\lambda v - 2\lambda\tau\rho + 3\gamma^2 - 8c_n\gamma - \gamma c_r + \gamma s - 2\rho c_r + 2\rho s + 8\lambda v}{4\gamma^2 - 17\gamma + 16} \\ & + \frac{-11\gamma + 8c_n + 2c_r - 2s + 8}{4\gamma^2 - 17\gamma + 16} \end{aligned}$$

$$\begin{aligned} w_r^{110} = & -\frac{1}{4\gamma^2\rho - 4\gamma^2 - 17\gamma\rho + 17\gamma + 16\rho - 16} \left(-2\gamma^2\rho\lambda\tau - 2\gamma^2\rho c_r + 2\gamma^2\rho s + 4\gamma^2\lambda\tau + \gamma^2\lambda v \right. \\ & \left. + 8\gamma\rho\lambda\tau + 2\gamma^3 - \gamma^2c_n + 2\gamma^2c_r - 2\gamma^2s + 8\gamma\rho c_r - 8\gamma\rho s - 17\gamma\lambda\tau - 2\gamma\lambda v - 8\lambda\tau\rho - 8\gamma^2 + 2c_n\gamma \right. \\ & \left. - 8\gamma c_r + 8\gamma s \right) \end{aligned}$$

Substituting the optimal wholesale prices above into the retail price and demand functions, then we can obtain the optimal profit of manufacturer and remanufacturer respectively:

$$\begin{aligned} \max \Pi_M^{110} = & \left(1 - \frac{\gamma^2 + \left(\lambda v - w_n^{110} - 3 \right) \gamma + \left(\tau - 2v \right) \lambda + 2w_n^{110} + 2 + \left(-1 + \rho \right) w_r^{110}}{(-1 + \gamma)(\gamma - 4)} \right) \left(w_n^{110} - c_n \right), \\ \max \Pi_{RM}^{110} = & \frac{\left(-1 + \lambda \right) \left(\left(\lambda v + 1 \right) \gamma^2 + \left(\left(-1 + \rho \right) w_r^{110} - 3\lambda v - w_n^{110} - 1 \right) \gamma + \left(-2\rho + 2 \right) w_r^{110} + 2\lambda\tau \right)}{\gamma(-1 + \gamma)(\gamma - 4)} \left(w_r^{110} - c_r + s \right) \\ & - \frac{\lambda \left(\left(\lambda v - v + 1 \right) \gamma^2 + \left(\left(-1 + \rho \right) w_r^{110} - 3\lambda v + \tau + 4v - w_n^{110} - 1 \right) \gamma + \left(-2\rho + 2 \right) w_r^{110} + 2\tau \left(\lambda - 2 \right) \right)}{\gamma(-1 + \gamma)(\gamma - 4)} \left(w_r^{110} - c_r - \tau + s \right) \end{aligned}$$

Proof of Theorem 5.

Since $\frac{\partial^2 \Pi_{R1}^{011}(p_n)}{\partial p_n^2} = -\frac{2}{1-\gamma} < 0$, $\frac{\partial^2 \Pi_{R2}^{011}(p_r)}{\partial p_r^2} = -\frac{2}{1-\gamma} - \frac{2}{\gamma} < 0$, Π_{R1}^{011} and Π_{R2}^{011} are strictly concave in p_n and p_r respectively. We can identify the optimal product prices by solving the first-order conditions of $\Pi_{R1}(p_n)$ and $\Pi_{R2}(p_r)$:

$$p_n^{011} = \frac{\gamma\lambda v + \lambda\tau - 2\lambda v + 2\gamma - 2w_n^{011} - w_r^{011} - 2}{\gamma - 4} \quad (\text{A21})$$

$$p_r^{011} = \frac{\lambda\tau\gamma + \gamma\lambda v + \gamma^2 - w_n^{011}\gamma - 2\lambda\tau - \gamma - 2w_r^{011}}{\gamma - 4} \quad (\text{A22})$$

Substituting the optimal price values from Eqs. (A21) and (A22) into the demand functions of q_n and q_r , we have

$$q_n^{011} = \frac{\left(-\lambda v + w_n^{011} - 2 \right) \gamma + \left(-\tau + 2v \right) \lambda - 2w_n^{011} + w_r^{011} + 2}{(\gamma - 4)(\gamma - 1)} \quad (\text{A23})$$

$$q_r^{011} = \frac{-\gamma^2 + \left(\left(-\tau - v \right) \lambda + w_n^{011} + w_r^{011} + 1 \right) \gamma + 2\lambda\tau - 2w_r^{011}}{(\gamma - 4)(\gamma - 1)\gamma} \quad (\text{A24})$$

$$q_r^{a^{011}} = \frac{\left(\left(\lambda v + 1 \right) \gamma^2 + \left(-3\lambda v - w_n^{011} - w_r^{011} - 1 \right) \gamma + 2\lambda\tau + 2w_r^{011} \right) \left(\lambda - 1 \right)}{(\gamma - 4)(\gamma - 1)\gamma} \quad (\text{A25})$$

$$q_r^{b^{011}} = \frac{-\gamma^2 + \left(\left(-\tau - v \right) \lambda + w_n^{011} + w_r^{011} + 1 \right) \gamma + 2\lambda\tau - 2w_r^{011}}{(\gamma - 4)(\gamma - 1)\gamma} \quad (\text{A26})$$

The profits of manufacturer and remanufacturer can be written as the following two formulas respectively:

$$\begin{aligned} \max \Pi_M^{011}(w_n, w_r) &= q_n^{011}(w_n - c_n) \\ \max \Pi_{RM}^{011}(w_n, w_r) &= q_r^{a^{011}}(w_r - c_r) + q_r^{b^{011}}(w_r - c_r - \tau) - \left(c_r q_r^{011} - B \right) \left(1 + r_B \right) \end{aligned}$$

Since $\frac{\partial^2 \Pi_M^{011}}{\partial w_n^2} = -\frac{2(2-\gamma)}{(\gamma-1)(\gamma-4)} < 0$ and $\frac{\partial^2 \Pi_{RM}^{011}}{\partial w_r^2} = \frac{2\gamma-4}{(\gamma-4)(\gamma-1)\gamma} < 0$, Π_{RM}^{011} and Π_M^{011} are strictly concave in w_r and w_n respectively. We can obtain the optimal wholesale prices of the remanufactured and new products by solving:

$$\frac{\partial \Pi_M^{011}}{\partial w_n} = -\frac{(\lambda v - w_n + 2)\gamma + 2w_n + (\tau - 2v)\lambda - w_r - 2 + (c_n - w_n)(2 - \gamma)}{(\gamma - 4)(\gamma - 1)} = 0 \quad (\text{A27})$$

$$\frac{\partial \Pi_{RM}^{011}}{\partial w_r} = \frac{-\gamma^2 + ((-2\tau - v)\lambda - (r_B + 2)c_r + 2w_r + w_n + 1)\gamma + 4\lambda\tau + 2(r_B + 2)c_r - 4w_r}{(\gamma - 4)(\gamma - 1)\gamma} = 0 \quad (\text{A28})$$

Combining Eqs. (A27) and (A28), we find

$$w_n^{011} = \frac{2\gamma^2\lambda v + 2\gamma^2c_n - 9\gamma\lambda v - \gamma c_r r_B + 3\gamma^2 - 8\gamma c_n - 2\gamma c_r + 8\lambda v + 2c_r r_B - 11\gamma + 8c_n + 4c_r + 8}{4\gamma^2 - 17\gamma + 16}$$

$$\begin{aligned} w_r^{011} &= \frac{4\gamma^2\lambda\tau + \gamma^2\lambda v + 2\gamma^2c_r r_B + 2\gamma^3 - \gamma^2c_n + 4\gamma^2c_r - 17\lambda\tau v - 2\gamma\lambda v - 8\gamma c_r r_B - 8\gamma^2 + 2\gamma c_n}{4\gamma^2 - 17\gamma + 16} \\ &\quad + \frac{-16\gamma c_r + 16\lambda\tau + 8c_r r_B + 6\gamma + 16c_r}{4\gamma^2 - 17\gamma + 16} \end{aligned}$$

Substituting the optimal wholesale prices above into Eqs. (A21)–(A26), we have the optimal retail price and the optimal product demand. Then we can obtain the optimal profit of manufacturer and remanufacturer respectively:

$$\max \Pi_M^{011} = \frac{\left(-\lambda v + w_n^{011} - 2 \right) \gamma + \left(-\tau + 2v \right) \lambda - 2w_n^{011} + w_r^{011} + 2}{(\gamma - 4)(\gamma - 1)} \left(w_n^{011} - c_n \right)$$

$$\begin{aligned} \max \Pi_{RM}^{011} &= \frac{\left(\left(\lambda v + 1 \right) \gamma^2 + \left(-3\lambda v - w_n^{011} - w_r^{011} - 1 \right) \gamma + 2\lambda\tau + 2w_r^{011} \right) \left(\lambda - 1 \right)}{(\gamma - 4)(\gamma - 1)\gamma} \left(w_r^{011} - c_r \right) \\ &\quad + \frac{-\gamma^2 + \left(\left(-\tau - v \right) \lambda + w_n^{011} + w_r^{011} + 1 \right) \gamma + 2\lambda\tau - 2w_r^{011}}{(\gamma - 4)(\gamma - 1)\gamma} \left(w_r^{011} - c_r - \tau \right) \\ &\quad - \left(c_r \frac{-\gamma^2 + \left(\left(-\tau - v \right) \lambda + w_n^{011} + w_r^{011} + 1 \right) \gamma + 2\lambda\tau - 2w_r^{011}}{(\gamma - 4)(\gamma - 1)\gamma} - B \right) \left(1 + r_B \right). \end{aligned}$$

Proof of Theorem 6.

Since $\frac{\partial^2 \Pi_{R1}^{012}}{\partial p_n^2} = -\frac{2}{1-\gamma} < 0$ and $\frac{\partial^2 \Pi_{R2}^{012}}{\partial p_r^2} = -\frac{2}{1-\gamma} - \frac{2}{\gamma} < 0$, Π_{R1}^{012} and Π_{R2}^{012} are strictly concave in p_n and p_r respectively. We can identify the optimal product prices by solving the first-order conditions of $\Pi_{R1}(p_n)$ and $\Pi_{R2}(p_r)$:

$$p_n^{012} = \frac{\gamma\lambda v + \lambda\tau - 2\lambda v + c_r r_R + 2a + c_r - 2w_n^{012} - w_r^{012} - 2}{\gamma - 4} \quad (\text{A29})$$

$$p_r^{012} = \frac{\lambda\tau\gamma + \gamma\lambda v + \gamma^2 - w_n^{012}\gamma - 2\lambda\tau + 2c_r r_R - \gamma + 2c_r - 2w_r^{012}}{\gamma - 4} \quad (\text{A30})$$

Substituting the optimal price values from Eqs. (A29) and (A30) into the demand functions of q_n and q_r , we obtain the optimal quantities of the two products:

$$q_r^{012} = \frac{-\gamma^2 + \left((-r_R - 1)c_r + (-\tau - v)\lambda + w_n^{012} + w_r^{012} + 1 \right)\gamma + (2r_R + 2)c_r + 2\lambda\tau - 2w_r^{012}}{(\gamma - 4)(-1 + \gamma)\gamma} \quad (A31)$$

$$q_n^{012} = \frac{(-\lambda\tau + w_n^{012} - 2)\gamma + (3\tau - 2v)\lambda + (-r_R - 1)c_r - 2w_n^{012} + w_r^{012} + 2}{(\gamma - 4)(-1 + \gamma)} \quad (A32)$$

$$q_r^{012} = \frac{(\lambda v + 1)\gamma^2 + (c_r(1 + r_R) - 3\lambda v - w_n^{012} - w_r^{012} - 1)\gamma + (2 - 2r_R)c_r + 2\lambda\tau + 2w_r^{012}}{(\gamma - 4)\gamma} \quad (A33)$$

$$q_r^{012} = \frac{(-\lambda v - 1)\gamma^3 + (4\lambda v - (r_R - 1)c_r + w_n^{012} + w_r^{012} + 1)\gamma^2 + \left((-3\tau - 4v)\lambda + (2r_R + 2)c_r - 2w_r^{012} \right)\gamma + 4\lambda\tau}{(\gamma - 4)(\gamma - 1)\gamma} \quad (A34)$$

The profits of manufacturer and remanufacturer can be written as the following two formulas respectively:

$$\max \Pi_M^{012}(w_n, w_r) = q_n^{012}(w_n - c_n)$$

$$\max \Pi_{RM}^{012}(w_n, w_r) = q_r^{012}(w_r - c_r) + q_r^{012}(w_r - c_r - \tau) - (c_r q_r^{012} - B)(1 + r_R)$$

Since $\frac{\partial^2 \Pi_M^{012}}{\partial w_n^2} = \frac{2(\gamma - 2)}{(\gamma - 4)(\gamma - 1)} < 0$ and $\frac{\partial^2 \Pi_{RM}^{012}}{\partial w_r^2} = \frac{2(\gamma - 2)}{(\gamma - 4)(\gamma - 1)\gamma} < 0$, Π_{RM}^{012} and Π_M^{012} are strictly concave in w_r and w_n respectively. We can obtain the optimal wholesale prices of the remanufactured and new products by solving:

$$\frac{\partial \Pi_M^{012}}{\partial w_n} = \frac{(\gamma - 2)(w_n - c_n) + (w_n - 2 - \lambda\tau)\gamma + (3\tau - 2v)\lambda + (-r_R - 1)c_r - 2w_n + w_r + 2}{(\gamma - 4)(\gamma - 1)} = 0 \quad (A35)$$

$$\frac{\partial \Pi_{RM}^{012}}{\partial w_r} = \frac{(-\tau - 1)\gamma^2 + ((-2r_R - 3)c_r + (2 - \lambda)\tau + 2w_r - \lambda v + w_n + 1)\gamma + (4r_R + 6)c_r + 2\lambda\tau - 4w_r}{(\gamma - 4)(\gamma - 1)\gamma} = 0 \quad (A36)$$

Combining Eqs. (A35) and (A36), we find

$$w_n^{012} = \frac{2\gamma^2\lambda\tau + 2\gamma^2c_n - \gamma^2\tau - 11\lambda\tau\gamma + 3\gamma\lambda v + 3\gamma^2 - 8\gamma c_n - \gamma c_r + 2\gamma\tau + 14\lambda\tau - 8\lambda v - 11\gamma + 8c_n + 2c_r + 8}{4\gamma^2 - 17\gamma + 16}$$

$$w_r^{012} = \frac{1}{4\gamma^2 - 17\gamma + 16} \left(2\gamma^3\tau + \gamma^2\lambda\tau + 2\gamma^2\lambda v + 4\gamma^2c_r r_R + 2\gamma^3 - \gamma^2c_n + 6\gamma^2c_r - 8\gamma^2\tau - 5\lambda\tau\gamma - 6\gamma\lambda v - 17\gamma c_r r_R - 8\gamma^2 + 2\gamma c_n - 25\gamma c_r + 8\gamma\tau + 8\lambda\tau + 16c_r r_R + 6\gamma + 24c_r \right)$$

Substituting the optimal wholesale prices above into Eqs. (A29)–(A34), we have the optimal retail price and the optimal product demand. Then we can obtain the optimal profit of manufacturer and remanufacturer respectively:

$$\max \Pi_M^{012} = \frac{(-\lambda\tau + w_n^{012} - 2)\gamma + (3\tau - 2v)\lambda + (-r_R - 1)c_r - 2w_n^{012} + w_r^{012} + 2}{(\gamma - 4)(-1 + \gamma)} \left(w_n^{012} - c_n \right)$$

$$\begin{aligned} \max \Pi_{RM}^{012} = & \frac{(\lambda v + 1)\gamma^2 + (c_r(1 + r_R) - 3\lambda v - w_n^{012} - w_r^{012} - 1)\gamma + (2 - 2r_R)c_r + 2\lambda\tau + 2w_r^{012}}{(\gamma - 4)\gamma} (w_r^{012} - c_r) \\ & + \frac{(-\lambda v - 1)\gamma^3 + (4\lambda v - (r_R - 1)c_r + w_n^{012} + w_r^{012} + 1)\gamma^2 + ((-3\tau - 4v)\lambda + (2r_R + 2)c_r - 2w_r^{012})\gamma + 4\lambda\tau}{(\gamma - 4)(\gamma - 1)\gamma} (w_r^{012} - c_r - \tau) \\ & + \frac{(2r_R + 2)c_r - 2w_r^{012}}{(\gamma - 4)(\gamma - 1)\gamma} (w_r^{012} - c_r - \tau) \\ & - (c_r - \gamma^2 + ((-r_R - 1)c_r + (-\tau - v)\lambda + w_n^{012} + w_r^{012} + 1)\gamma + (2r_R + 2)c_r + 2\lambda\tau - 2w_r^{012}) / ((\gamma - 4)(-1 + \gamma)\gamma) - B)(1 + r_R) \end{aligned}$$

Then we can get the Theorem 6.

Proof of Theorem 7.

Same as Theorem 3, we can get the optimal retail price and quantities. Taking the first derivative of Π_{R1}^{11} and Π_{R2}^{11} with respect to p_n and p_r , then substituting the optimal price values into the demand functions of q_n and q_r , we have:

$$p_n^{11} = \frac{\gamma\lambda v + \rho w_r^{11} + \lambda\tau - 2\lambda v + 2\gamma - 2w_n^{11} - w_r^{11} - 2}{\gamma - 4}$$

$$p_r^{111} = -\frac{\gamma\lambda\tau + \gamma\lambda v + \gamma^2 - \gamma w_n^{111} + 2\rho w_r^{111} - 2\lambda\tau - \gamma - 2w_r^{111}}{\gamma\rho - \gamma - 4\rho + 4}$$

$$q_n^{111} = 1 - \frac{\gamma^2 + (\lambda v - w_n^{111} - 3)\gamma + (\tau - 2v)\lambda + 2w_n^{111} + 2 + (-1 + \rho)w_r^{111}}{(-1 + \gamma)(\gamma - 4)}, \quad q_r^{111} = \frac{(-1 + \lambda)((\lambda v + 1)\gamma^2 + ((-1 + \rho)w_r^{111} - 3\lambda v - w_n^{111} - 1)\gamma + (-2\rho + 2)w_r^{111} + 2\lambda\tau)}{\gamma(-1 + \gamma)(\gamma - 4)},$$

$$q_r^{b111} = -\frac{\lambda((\gamma v - v + 1)\gamma^2 + ((\rho - 1)w_r^{111} - 3\lambda v + \tau + 4v - w_n^{111} - 1)\gamma + (2 - 2\rho)w_r^{111} + 2\tau(\lambda - 2))}{(\gamma - 4)(\gamma - 1)\gamma}, \quad q_r^{111} = -\frac{\gamma^2 + (\rho w_r^{111} + (\tau + v)\lambda - w_n^{111} - 1)\gamma - 2\rho w_r^{111} - 2\lambda\tau + 2w_r^{111}}{(\gamma - 4)(\gamma - 1)\gamma}$$

The profits of manufacturer and remanufacturer can be written as the following two formulas respectively:

$$\max \Pi_M^{111}(w_n, w_r) = q_n^{112}(w_n - c_n)$$

$$\max \Pi_{RM}^{111}(w_n, w_r) = q_r^{a111}(w_r - c_r + s) + q_r^{b111}(w_r - c_r - \tau + s) - (c_r q_r^{111} - B) \left(1 + r_R \right)$$

Since $\frac{\partial^2 \Pi_M^{111}}{\partial w_n^2} = -\frac{2(2-\gamma)}{(\gamma-1)(\gamma-4)} < 0$ and $\frac{\partial^2 \Pi_{RM}^{111}}{\partial w_r^2} = -\frac{2(\rho-1)(\rho-2)}{(\gamma-4)(\gamma-1)\gamma} < 0$, Π_M^{111} and Π_{RM}^{111} are strictly concave in w_n and w_r respectively. We obtain the optimal wholesale prices of the remanufactured and new products by solving

$$\frac{\partial \Pi_M^{111}(w_n, w_r)}{\partial w_n} = -\frac{(\lambda v - w_n + 2)\gamma + 2w_n + (\tau - 2v)\lambda - 2 + (\rho - 1)w_r + (c_n - w_n)(2 - \gamma)}{(\gamma - 1)(\gamma - 4)} = 0 \quad (A37)$$

$$\frac{\partial \Pi_{RM}^{111}(w_n, w_r)}{\partial w_n} = \frac{((\lambda v + 1)\gamma^2 + ((\rho - 1)w_r - 3\lambda v - w_n - 1)\gamma + 2\lambda\tau - (2\rho - 2)w_r)(\lambda - 1)}{(\gamma - 4)(\gamma - 1)\gamma}$$

$$- \frac{((\lambda v - v + 1)\gamma^2 + ((\rho - 1)w_r - 3\lambda v + \tau + 4v - w_n - 1)\gamma + 2\lambda\tau + (2 - 2\rho)w_r - 4\tau)\lambda}{(\gamma - 4)(\gamma - 1)\gamma}$$

$$+ \frac{((\rho - 1)\gamma + 2 - 2\rho)c_r(1 + r_B) + (w_r - c_r + s)((\rho - 1)\gamma + 2 - 2\rho)(\lambda - 1)}{(\gamma - 4)(\gamma - 1)\gamma} = 0 \quad (A38)$$

Combining Eq. (A37) and (A38), we find

$$w_n^{111} = \frac{1}{4\gamma^2 - 17\gamma + 16} \left(2\gamma^2\lambda v + \gamma\rho\lambda\tau + \gamma\rho c_r r_B + 2\gamma^2 c_n + 2\gamma\rho c_r - \gamma\rho s - 9\gamma\lambda v - \gamma c_r r_B - 2\rho\lambda\tau - 2\rho c_r r_B + 3\gamma^2 - 8\gamma c_n \right.$$

$$\left. - 2\gamma c_r + \gamma s - 4\rho c_r + 2\rho s + 8\lambda v + 2c_r r_B - 11\gamma + 8c_n + 4c_r - 2s + 8 \right)$$

$$w_r^{111} = \frac{1}{4\gamma^2\rho - 4\gamma^2 - 17\gamma\rho + 17\gamma + 16\rho - 16} \left(-2\gamma^2\rho\lambda\tau - 2\gamma^2\rho c_r r_B - 4\gamma^2\rho c_r + 2\gamma^2\rho s + 4\gamma^2\lambda\tau + \gamma^2\lambda v + 2\gamma^2 c_r r_B \right.$$

$$+ 8\gamma\rho\lambda\tau + 8\gamma\rho c_r r_B + 2\gamma^3 - \gamma^2 c_n + 4\gamma^2 c_r - 2\gamma^2 s + 16\gamma\rho c_r - 8\gamma\rho s - 17\gamma\lambda\tau - 2\gamma\lambda v - 8\gamma c_r r_B - 8\rho\lambda\tau - 8\rho c_r r_B - 8\gamma^2$$

$$\left. + 2c_n\gamma - 16\gamma c_r + 8\gamma s - 16\rho c_r + 8\gamma s - 16\rho c_r + 8\rho s \right)$$

Substituting the optimal wholesale prices above into the retail price and demand functions, then we can obtain the optimal profit of manufacturer and remanufacturer respectively:

$$\max \Pi_M^{111} = \left(1 - \frac{\gamma^2 + (\lambda v - w_n^{111} - 3)\gamma + (\tau - 2v)\lambda + 2w_n^{111} + 2 + (-1 + \rho)w_r^{111}}{(-1 + \gamma)(\gamma - 4)} \right) \left(w_n^{111} - c_n \right)$$

$$\max \Pi_{RM}^{111}(w_n, w_r) = \frac{(-1 + \lambda)((\lambda v + 1)\gamma^2 + ((-1 + \rho)w_r^{111} - 3\lambda v - w_n^{111} - 1)\gamma + (-2\rho + 2)w_r^{111} + 2\lambda\tau)}{\gamma(-1 + \gamma)(\gamma - 4)} \left(w_r - c_r + s \right)$$

$$- \frac{\lambda((\gamma v - v + 1)\gamma^2 + ((\rho - 1)w_r^{111} - 3\lambda v + \tau + 4v - w_n^{111} - 1)\gamma + (2 - 2\rho)w_r^{111} + 2\tau(\lambda - 2))}{(\gamma - 4)(\gamma - 1)\gamma}$$

$$\left(w_r - c_r - \tau + s \right) - \left(-c_r \frac{\gamma^2 + (\rho w_r^{111} + (\tau + v)\lambda - w_n^{111} - 1)\gamma - 2\rho w_r^{111} - 2\lambda\tau + 2w_r^{111}}{(\gamma - 4)(\gamma - 1)\gamma} - B \right) \left(1 + r_R \right)$$

Proof of Theorem 8.

Since $\frac{\partial^2 \Pi_{R1}^{112}}{\partial p_n^2} = -\frac{2}{1-\gamma} < 0$ and $\frac{\partial^2 \Pi_{R2}^{112}}{\partial p_r^2} = \frac{2(1-\rho)}{(\gamma-1)\gamma} < 0$, Π_{R1}^{112} and Π_{R2}^{112} are strictly concave in p_n and p_r respectively. We can identify the optimal product prices by solving the first-order conditions of $\Pi_{R1}(p_n)$ and $\Pi_{R2}(p_r)$:

$$p_n^{112} = \frac{\gamma\lambda v - \rho c_r r_R - \rho c_r + \rho w_r^{112} + \lambda\tau - 2\lambda v + c_r r_R + 2\gamma + c_r - 2w_n^{112} - w_r^{112} - 2}{\gamma - 4}$$

$$p_r^{112} = -\frac{\gamma\lambda\tau + \gamma\lambda v - 2\rho c_r r_R + \gamma^2 - \gamma w_n^{112} - 2\rho c_r + 2\rho w_r^{112} - 2\lambda\tau + 2c_r r_R - \gamma + 2c_r - 2w_r^{112}}{\gamma\rho - \gamma - 4\rho + 4}$$

Substituting the optimal price values into the demand functions of q_n and q_r , we obtain the optimal quantities of the two products:

$$q_n^{112} = 1 - \frac{\gamma^2 + (\lambda v - w_n^{112} - 3)\gamma - (\rho - 1)c_r(r_R + 1) + \rho w_r^{112} + (\tau - 2v)\lambda + 2w_n^{112} - w_r^{112} + 2}{(\gamma - 4)(-1 + \gamma)}$$

$$q_r^{112} = \frac{-\gamma^2 + \left((\rho - 1)c_r(r_R + 1) - \rho w_r^{112} + w_r^{112} + (-\tau - v)\lambda + w_n^{112} + 1\right)\gamma - 2(\rho - 1)c_r(r_R + 1) + (2\rho - 2)w_r^{112} + 2\lambda\tau}{(\gamma - 4)(-1 + \gamma)\gamma}$$

$$q_r^{a112} = \frac{\left((\lambda v + 1)\gamma^2 + (r_R + 1)(1 - \rho)c_r + \rho w_r - 3\lambda v - w_n^{112} - w_r^{112} - 1\right)\gamma + 2(r_R + 1)(\rho - 1)c_r - 2\rho w_r^{112} + 2\lambda\tau + 2w_r^{112}}{\gamma(\gamma - 1)(\gamma - 4)} \left((\lambda - 1)\right)$$

$$q_r^{b112} = -\frac{\lambda\left((\lambda v - v + 1)\gamma^2 + (r_R + 1)(1 - \rho)c_r + \rho w_r - 3\lambda v + \tau + 4v - w_n^{112} - w_r^{112} - 1\right)\gamma + 2(r_R + 1)(\rho - 1)c_r}{\gamma(\gamma - 1)(\gamma - 4)} + \frac{-2\rho w_r^{112} + 2\lambda\tau - 4\tau + 2w_r^{112}}{\gamma(\gamma - 1)(\gamma - 4)}$$

The profits of manufacturer and remanufacturer can be written as the following two formulas respectively:

$$\max \Pi_M^{112}(w_n, w_r) = q_n^{112}(w_n - c_n) \quad (A39)$$

$$\max \Pi_{RM}^{112}(w_n, w_r) = q_r^{a112}(w_r - c_r + s) + q_r^{b112}(w_r - c_r - \tau + s) - (c_r q_r^{112} - B)(1 + r_R) \quad (A40)$$

Since $\frac{\partial^2 \Pi_M^{112}}{\partial w_n^2} = -\frac{2(2-\gamma)}{(\gamma-1)(\gamma-4)} < 0$ and $\frac{\partial^2 \Pi_{RM}^{112}}{\partial w_r^2} = -\frac{2(\rho-1)(\gamma-2)}{\gamma(\gamma-1)(\gamma-4)} < 0$, Π_M^{112} and Π_{RM}^{112} are strictly concave in w_n and w_r respectively. We obtain the optimal wholesale prices of the remanufactured and new products by solving (A39) and (A40).

$$w_n^{112} = \frac{2\gamma^2\lambda v + \gamma\rho\lambda\tau + 2\gamma^2c_n + \gamma\rho c_r - \gamma\rho s - 9\gamma\lambda v - 2\rho\lambda\tau + 3\gamma^2 - 8c_n\gamma - \gamma c_r + \gamma s - 2\rho c_r + 2\rho s + 8\lambda v - 11\gamma + 8c_n}{4\gamma^2 - 17\gamma + 16}$$

$$w_r^{112} = -\frac{1}{4\gamma^2\rho - 4\gamma^2 - 17\rho\gamma + 17\gamma + 16\rho - 16} \left(-2\gamma^2\rho\lambda\tau - 4\gamma^2\rho c_r r_R - 6\gamma^2\rho c_r + 2\gamma^2\rho s + 4\gamma^2\lambda\tau + \gamma^2\lambda v + 4\gamma^2c_r r_R + 8\gamma\rho\lambda\tau + 17\gamma\rho c_r r_R + 2\gamma^3 - \gamma^2c_n + 6\gamma^2c_r - 2\gamma^2s + 25\gamma\rho c_r - 8\gamma\rho s - 17\gamma\lambda\tau - 2\gamma\lambda v - 17\gamma c_r r_R - 8\rho\lambda\tau - 16\rho c_r r_R - 8\gamma^2 + 2\gamma c_n - 25\gamma c_r + 8\gamma s - 24\rho c_r + 8\rho s + 16\lambda\tau + 16c_r r_R + 6\gamma + 24c_r - 8s \right)$$

Substituting the optimal wholesale prices above into the retail price and demand functions, then we can obtain the optimal profits of manufacturer and remanufacturer respectively:

$$\begin{aligned} \Pi_M^{112} &= (1 - \frac{\gamma^2 + (\lambda v - w_n^{112} - 3)\gamma - (\rho - 1)c_r(r_R + 1) + \rho w_r^{112} + (\tau - 2v)\lambda + 2w_n^{112} - w_r^{112} + 2}{(\gamma - 4)(-1 + \gamma)})(w_n^{112} - c_n) \\ \Pi_{RM}^{112} &= \frac{((\lambda v + 1)\gamma^2 + (r_R + 1)(1 - \rho)c_r + \rho w_r - 3\lambda v - w_n^{112} - w_r^{112} - 1)\gamma + 2(r_R + 1)(\rho - 1)c_r - 2\rho w_r^{112} + 2\lambda\tau + 2w_r^{112}}{\gamma(\gamma - 1)(\gamma - 4)} (w_r^{112} - c_r + s) \\ &\quad + \left(-\frac{\lambda((\lambda v - v + 1)\gamma^2 + (r_R + 1)(1 - \rho)c_r + \rho w_r - 3\lambda v + \tau + 4v - w_n^{112} - w_r^{112} - 1)\gamma + 2(r_R + 1)(\rho - 1)c_r}{\gamma(\gamma - 1)(\gamma - 4)} \right. \\ &\quad \left. + \frac{-2\rho w_r^{112} + 2\lambda\tau - 4\tau + 2w_r^{112}}{\gamma(\gamma - 1)(\gamma - 4)} \right) (w_r^{112} - c_r - \tau + s) \\ &\quad - (c_r \frac{-\gamma^2 + ((\rho - 1)c_r(r_R + 1) - \rho w_r^{112} + w_r^{112} + (-\tau - v)\lambda + w_n^{112} + 1)\gamma - 2(\rho - 1)c_r(r_R + 1) + (2\rho - 2)w_r^{112} + 2\lambda\tau}{(\gamma - 4)(-1 + \gamma)\gamma} - B)(1 + r_R) \end{aligned}$$

Then we can obtain the [Theorem 8](#).

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